

SKYG

SKYG SPEED 120B

USER MANUAL

REVISION C7B-1 · 2026-10-09



PACKAGE CONTENTS

Manual

SkyG Speed 120B user manual (this document).

STEP

Every printed part and the full assembly, for editing in any CAD program.

ArduPilot

Setup document and a starting .param file for the SpeedyBee F405 WING MINI.

Engineering

Design record, parts list (BOM.csv), test plan and verification report.

STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

Wiring

The wiring diagram as a separate large page.

Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

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Assembly guide

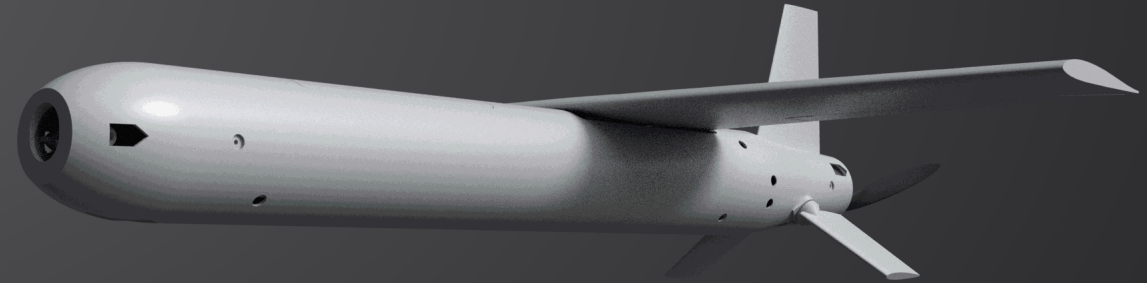
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SKYG SPEED 120B

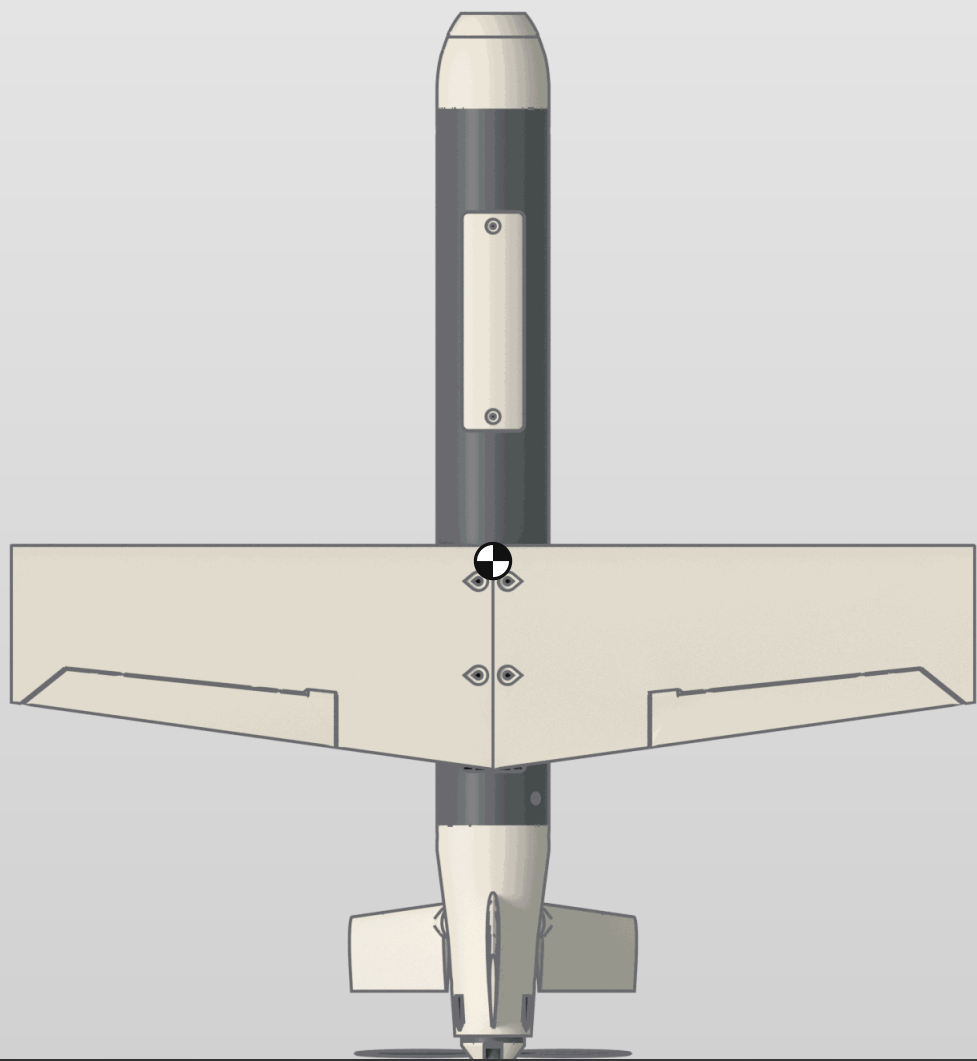
SkyG Speed 120B is a compact, hand-launched speed aircraft: a 3D-printed fuselage in three plug-together modules and a separate camera nose, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

The airframe is printed in foaming PLA Aero on a 0.4 mm nozzle; steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Six interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera or a DJI O4 Air Unit, each with an open or a clear-dome face.

It is built for straight-line speed: about 107 m/s finished with the planned rpm governor, and about 78 m/s until that governor is fitted, flown up through never-exceed stages.



GENERAL AIRCRAFT DATA



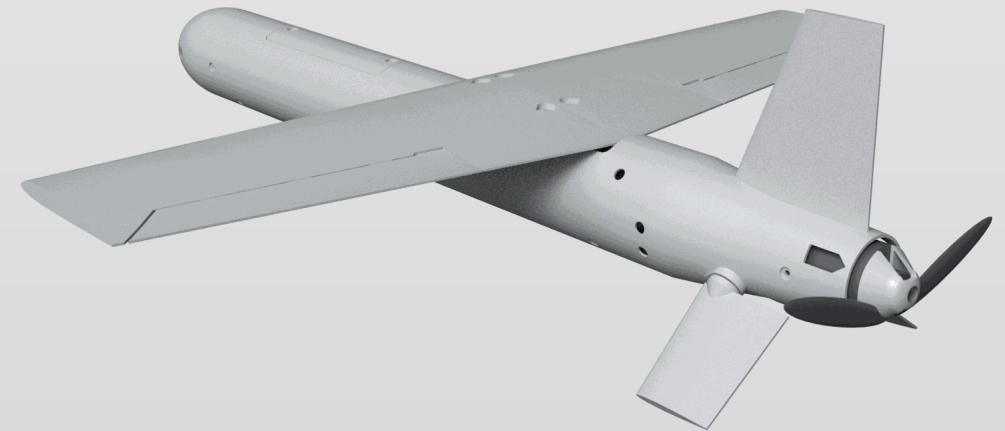
Wingspan	430 mm
Wing area	3.65 dm ²
Length	471 mm
Centre of gravity	7 mm behind the wing leading edge at the root
All-up weight	685 g (micro-camera nose)
Stall speed	13.0 m/s
Top speed (model)	106–107 m/s with the governor
Airfoil	MH45, ailerons 3° reflex
Root / tip chord	100 / 70 mm
MAC	86 mm
Aspect ratio	5.1
Wing loading	104 g/dm ²

AERODYNAMIC DESIGN

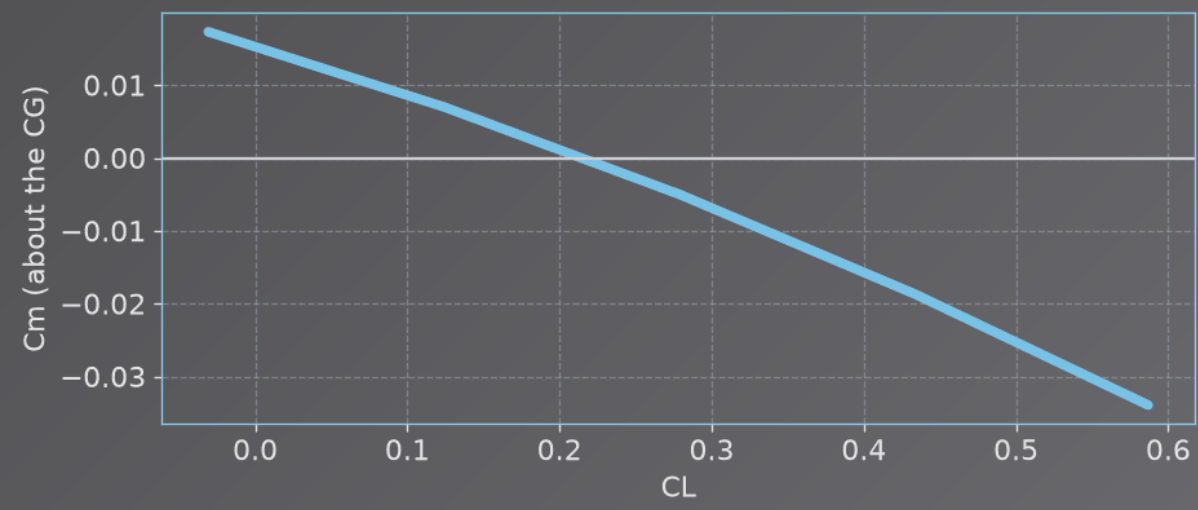
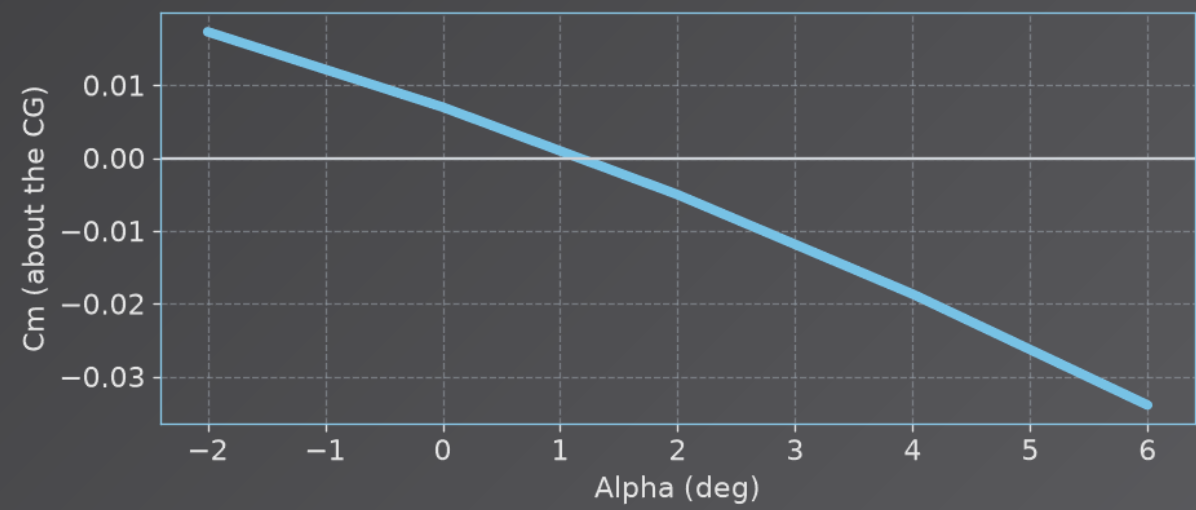
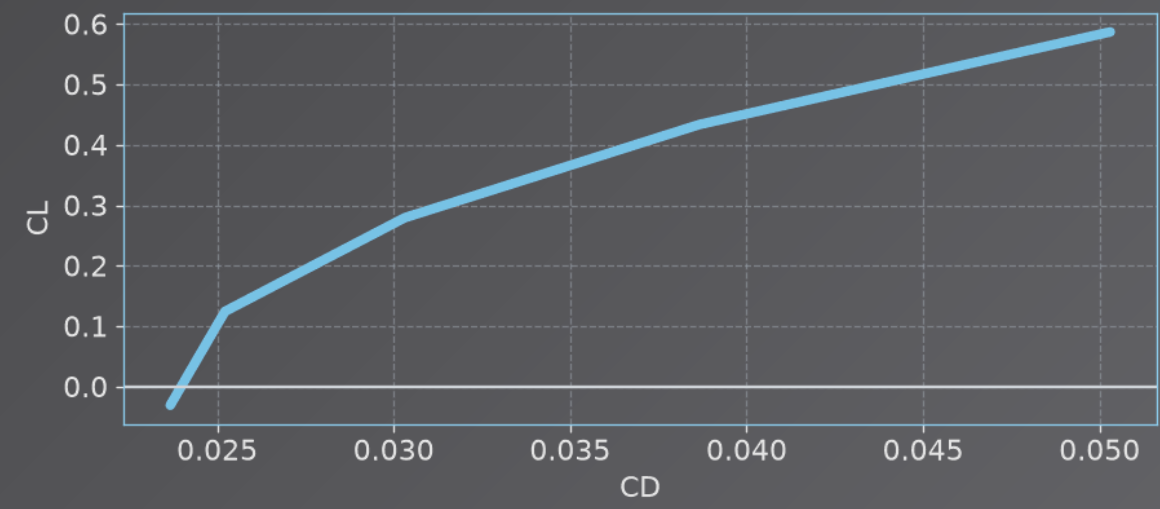
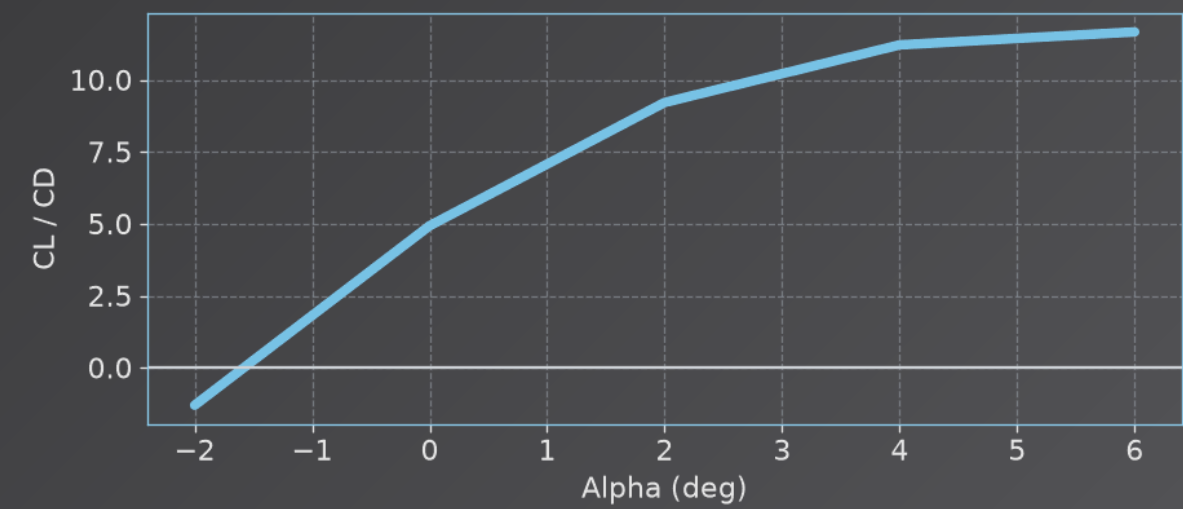
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 8 % of the mean chord, 6.5 % static margin ahead of the neutral point (x 258 mm from the nose datum): stable without a tail download penalty at speed.

Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 134 m/s, above 1.2 × the 90 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.



AERODYNAMIC CHARACTERISTICS



Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 11.7 at 6.0°; Cm about the design CG.

PERFORMANCE AND SPEED STAGES

Motor	T-Motor F50 2207 2150KV
Propeller	Gemfan 5x7.5E Vortex
Battery	BetaFPV LAVA II 3S 680 95C LiHV
Static thrust	1572 g at 65 A (thrust / weight 2.29)
Throttle cap (prop rpm rating)	70 %
Top speed	78–78 m/s now; 106–107 m/s with the governor
Full-throttle time	about 1.1 min on the four packs (bursts of ≤ 10 s)

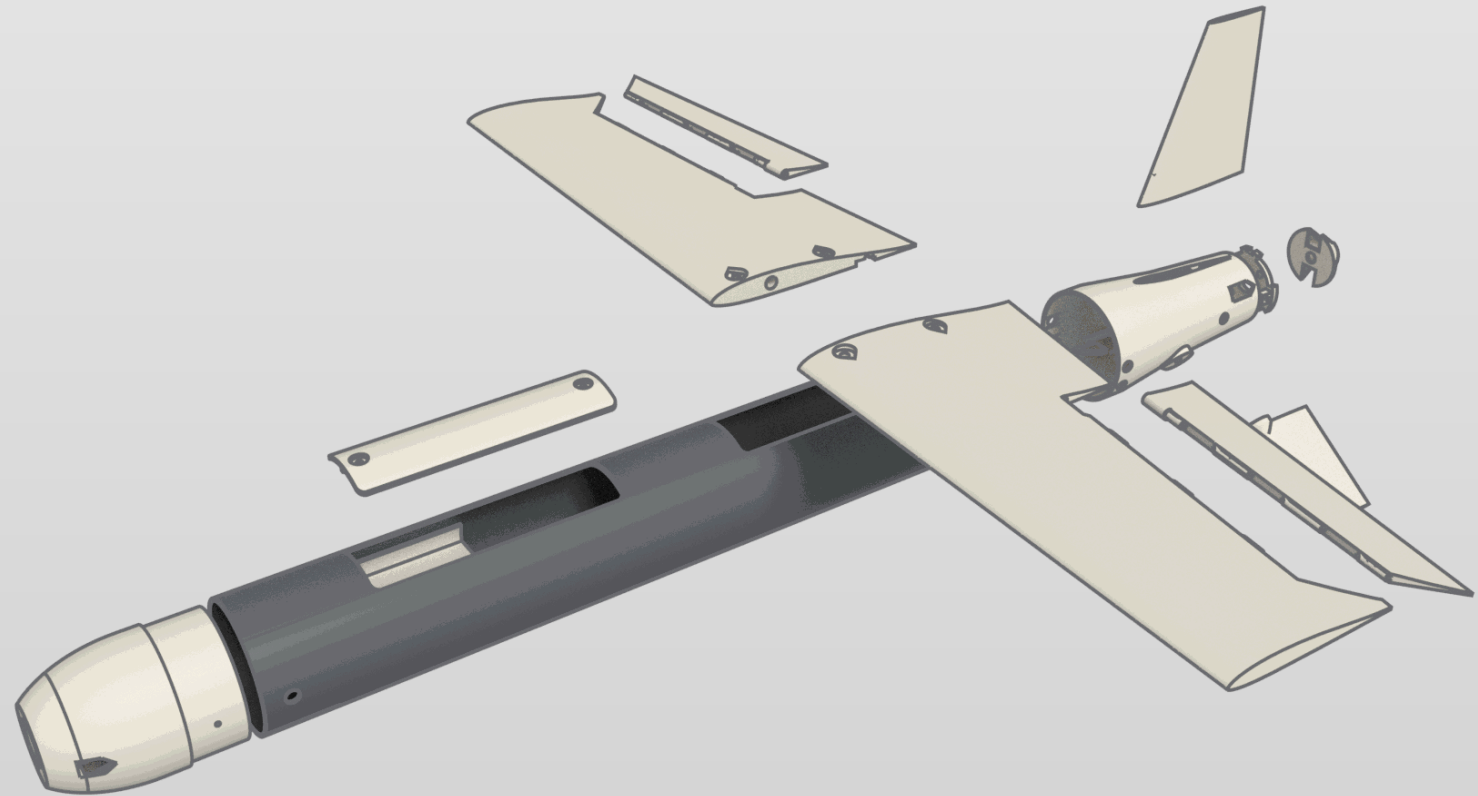
Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	23 %	23 m/s
35 m/s	26 %	28 m/s
40 m/s	29 %	32 m/s
45 m/s	33 %	37 m/s
50 m/s	36 %	41 m/s
55 m/s	40 %	45 m/s
60 m/s	44 %	50 m/s
65 m/s	47 %	53 m/s
70 m/s	51 %	58 m/s
75 m/s	55 %	62 m/s
80 m/s	59 %	66 m/s
85 m/s	63 %	70 m/s
90 m/s	67 %	74 m/s

No airspeed sensor: each stage is held by a throttle limit on a transmitter switch (LAUNCH = full cap for the throw and climb-out only, STAGE = the limit). Level runs only; speed from GPS two-way passes.

FILES



Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons and tail panels are separate files because they are mirror images. The interactive viewer in the package shows every part, purchased item and cable in place.



COMPARTMENT LAYOUT

BATTERY BAY

6S 1360 mAh as four 3S packs (two side-by-side pairs, one behind the other) on the tray under the wing; they stay in and are charged through the port under the hatch; balance position x 219–229

WING BAY

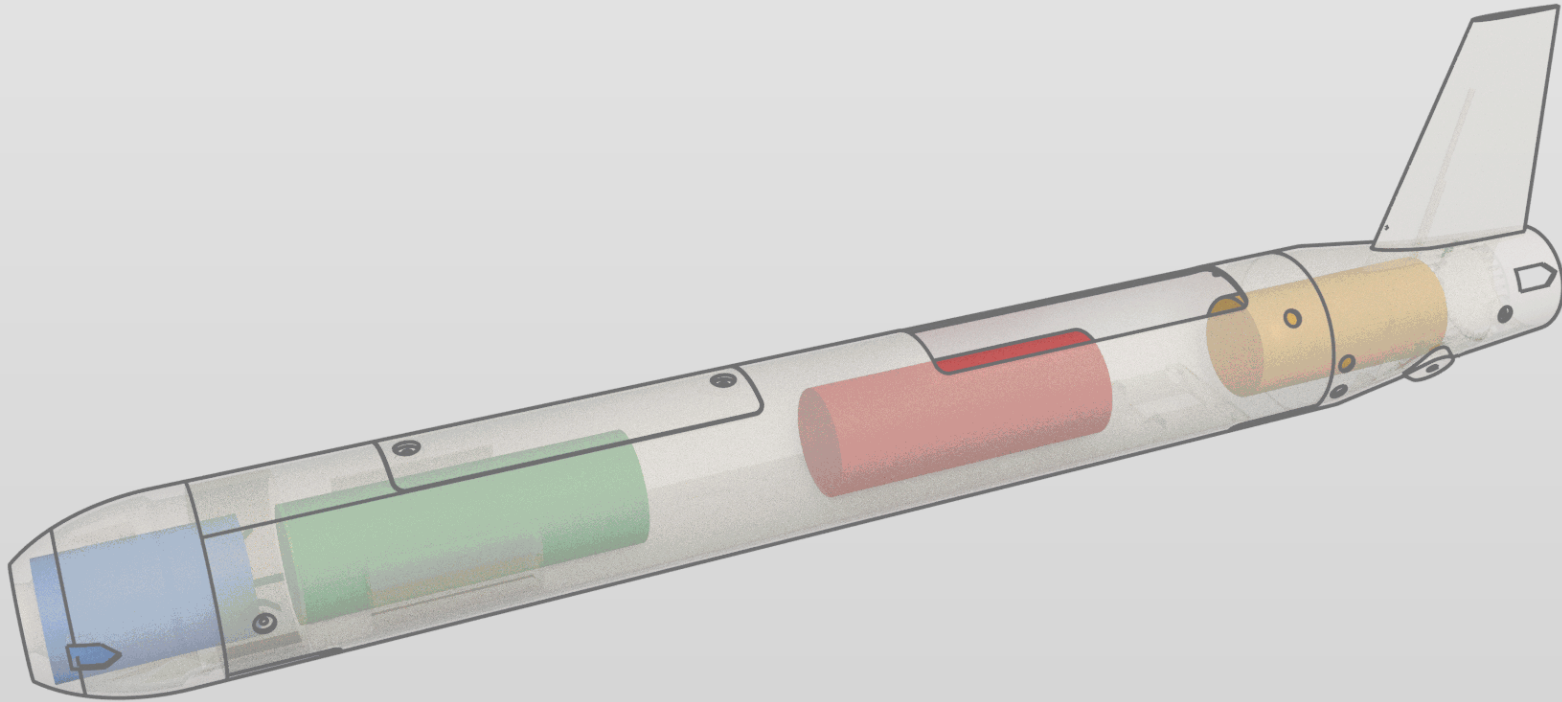
Flight controller stack with the ESC on top, receiver, aileron servo; reached with the wing off

TAIL MODULE

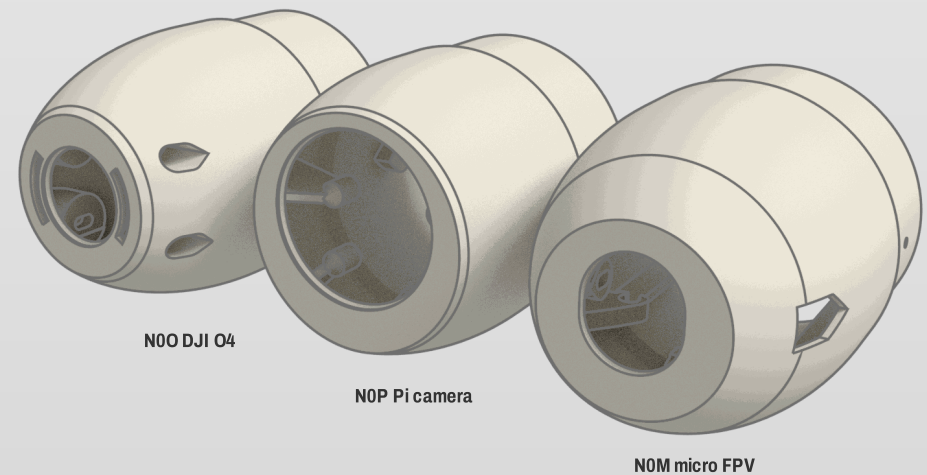
Two tail servos, pushrods, horns and the motor; plugs off at the MR30 and servo plugs

NOSE BAY

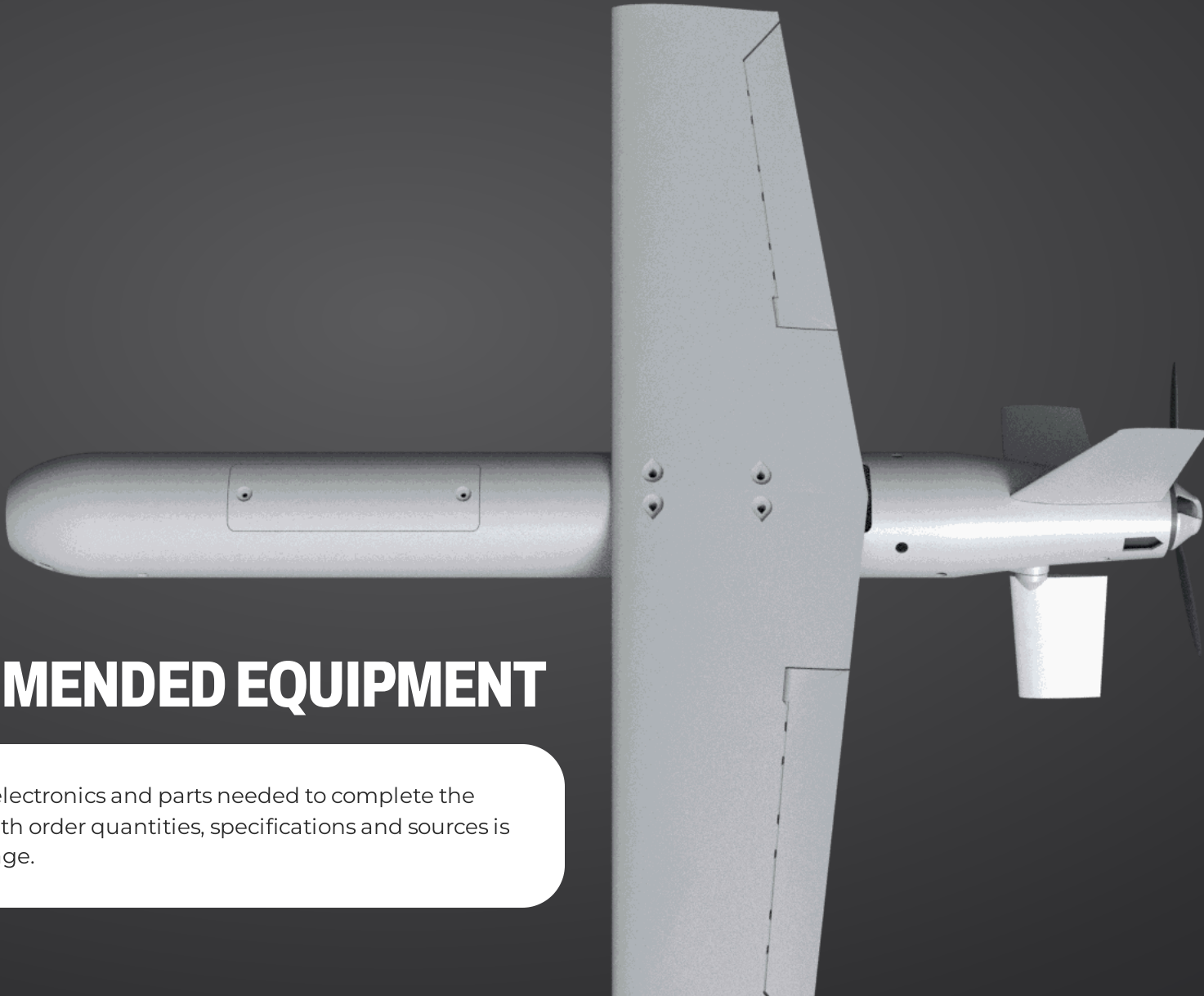
Camera and VTX (or the DJI O4 unit) in the removable nose



CAMERA NOSE VARIANTS



Nose	Camera	Battery centre for balance
N0M	19 mm micro FPV camera + 20 × 20 VTX	x 228 mm
N0P	Raspberry Pi Camera Module 3 (test option)	x 216 mm (at its stop + nose ballast)
N0MD	micro camera behind a 40 mm dome, nano VTX under the hatch	x 0 mm
N0PD	Pi camera behind a dome	x 0 mm
N0O	DJI O4 Air Unit (Lite), ducted	x 0 mm
N0OD	DJI O4 behind a dome	x 0 mm



RECOMMENDED EQUIPMENT



This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

RECOMMENDED ELECTRONICS

COMPONENT	MODEL / SPECIFICATION
Motor	T-Motor F50 2207 2150KV
Propeller (fixed)	Gemfan 5x7.5E Vortex
ESC	HDZero Halo 4in1 70A ESC AM32
Flight controller	SpeedyBee F405 WING MINI
GPS + compass	TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g
Servos	3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)
Flight battery	4 × BetaFPV LAVA II 3S 680 95C LiHV (2-pack)
Receiver	RadioMaster RP1 V2
FPV camera	Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2
Video transmitter + antenna	20x20 analog or digital VTX with U.FL antenna
DJI O4 Air Unit (Lite)	DJI O4 Air Unit

RECOMMENDED ACCESSORIES

ITEM	QTY
XT30 connector pairs	2
Carbon tube	1
Music wire 2.0 mm	2
Pushrods 1.2 mm with Z-bend	4
Pushrod keepers for 1.2 mm wire	4
Z-bend pliers, 1.1 mm drill, 1.2 mm drill (as a hand reamer), small flat file	0
Steel countersunk screw M3 x 16	2
M3 hex nut	4
M2 x 6 countersunk screws (joints)	4
Kapton / thin clear tape	as needed
Adhesives	as needed
Velcro strap + hook/loop	2
Heat shrink + silicone wire	as needed
Silver steel rod 2.5 mm	2
Brass tube 3 mm	2
MR30 connector pairs	1
Servo extension leads 300 mm	3
Motor screws M3	4
Steel rod 1.5 mm	2

ITEM	QTY
Steel cup washer M3 (countersunk finishing washer)	4
Steel cup washer M2 (countersunk finishing washer)	2
M2 x 6 countersunk screws (motor plate)	2
Servo strap / zip tie + foam tape	1
Steel rod 1.5 mm	2
Vent tape	1
15T servo arms	1
Dupont housings + crimps (1-pin, 4-pin)	3
Carbon-fibre tube 50 x 48 mm	320 mm
M2 x 6 countersunk screws (tube)	8
Structural epoxy	~3 g
Thin CA	~1 g
M2 x 6 countersunk screws (hatch)	2
XT30 series adapter (2 into 1)	2
XT30 parallel adapter (2 into 1)	1
Balance series adapter 2 x 3S to 6S	2
6S parallel balance board	1
Charge / arming port leads	1

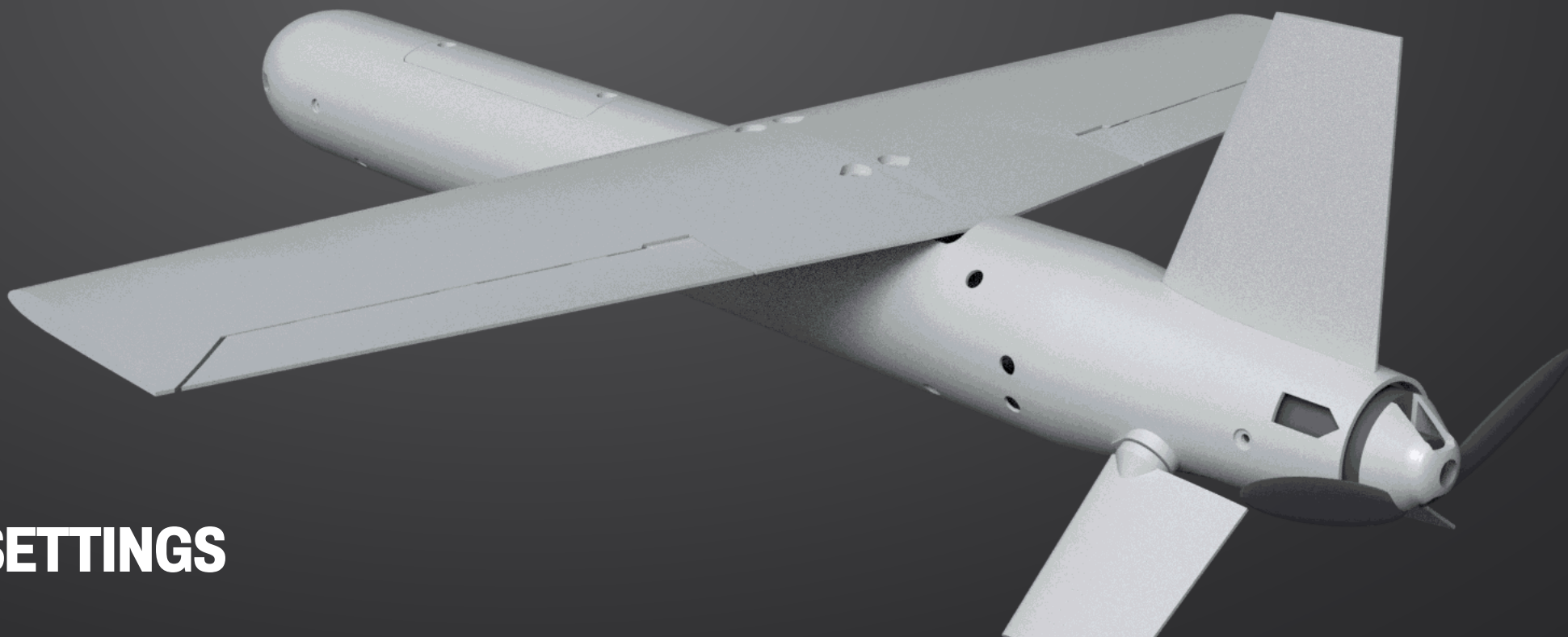


Motor and prop. T-Motor F50 2207 2150KV (6S-rated, 24 V; Ø27 x 29.5 mm incl. shaft, 28.56 g, 16 x 16 mm M3 with 4 mm thread depth, 5 mm shaft, M5 x 10.75 mm prop thread, 50 mOhm, 10 1.6 A @10 V; 54.8 A / 1273 W peak): the SkyG Speed 60 motor, same mount. Prop: Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre polyamide, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring; Robocombo stock). Gemfan publishes no rpm rating: each batch is spin-tested to 45,600 rpm (1.2 x the 38,000 rpm governor ceiling) behind a shield before flight. Pusher: the 5x7.5R, or the E mounted front face toward the motor with the motor reversed in the ESC; throttle-capped (no governor on the F405).

Throttle cap. The Gemfan 5 x 7.5E stays at or below 38,000 rpm. On a full 6S pack the unloaded prop would exceed it at full throttle, so the flight controller caps the throttle at 70 %. Never raise it, and inspect the blades after every thrust-stand run and flying day.

rpm governor (open item). A governor would hold the prop at 38,000 rpm and let it use that rpm in level flight. ArduPilot's governor script does not run on the F405 flight controller, so until an H7 flight controller (C7 uses the Matek H743-WLITE) is fitted the throttle cap above applies: about 78 m/s instead of 107 m/s.

Battery. Four BetaFPV LAVA II 3S 680 95C LiHV packs: each side-by-side pair in series (XT30 series adapter), the two pairs in parallel (XT30 parallel adapter), 6S 1360 mAh. They stay in the aircraft and are charged through the port under the hatch (XT30 + 6S balance plug; 6S LiPo mode, 4.20 V per cell, at most 1C, attended, on a non-flammable surface). The port's XT30 is also the arming plug. Full throttle in bursts of ≤ 10 s, then ≥ 30 s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 1572 g at 65 A; confirm on a thrust stand before the first flight.



PRINT SETTINGS

PRINT SETTINGS



Filament per part (no glass skin): **PLA-CF** wing halves (2 walls), **PETG-CF** ailerons, equipment tray, tail coupler, nut bars, hatch lips, duct inserts, motor plate and spinner; **PETG** camera noses, hatch cover and GPS carrier (radio windows: never carbon-filled filament there); **ASA** tail module and fin (motor heat; the fin carries the receiver antenna); PETG-CF tail panels and servo horns. 0.4 mm hardened-steel nozzle, 0.20 mm layers; every part fits both the H2S and the H2D build volume (340 × 320 × 325 mm).

GENERAL GUIDELINES

- Fuselage, nose and hatch shells: 2 walls, 30 % lightning infill (it only grows under top surfaces: at 0 % the solid bosses and bridges print hollow and their tops collapse).
- Wing halves: PLA-CF, 2 walls, 8 % gyroid. Ailerons: PETG-CF, 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid. Fin: 2 walls, 15 %.
- Filament per part: see the parts list (PLA-CF, PETG-CF, PETG, ASA, PLA). CF filaments need a hardened-steel nozzle; ASA an enclosed printer.
- Tall standing parts: 5 mm brim, slow outer wall, enclosure door closed.

SUPPORTS: OFF

Every part is designed to print without supports in the orientation shown, and checked on the slicer's own G-code with these settings: nothing is printed over air and no bridge is longer than 10 mm. Keep supports switched off and use the supplied .3mf files (they carry the settings). The downloads hold ONE set of Bambu Studio projects for the H2S and the H2D: no G-code inside (G-code only fits the printer it was sliced for), tree supports on (auto). Open, select your printer, slice, print. Check the top of the aileron's torque-rod bore on the first aileron.

PRINT SETTINGS

FIRST: MEASURE THE TUBE

Measure the carbon tube's bore at both ends before printing anything that fits inside it (the design uses 48.0 mm). If it differs by more than 0.1 mm, the inside parts must be rebuilt for your tube. Print the tail coupler first and try it in both ends of the tube: it must slide in with a light push.

COUPON TESTS

Print the camera nose and try its spigot in the tube's front; slide the tail module onto the coupler; print the tray and slide it the whole length of the tube before cutting any window. Print a 40 mm wing-root section in PLA-CF and check the spar fit.

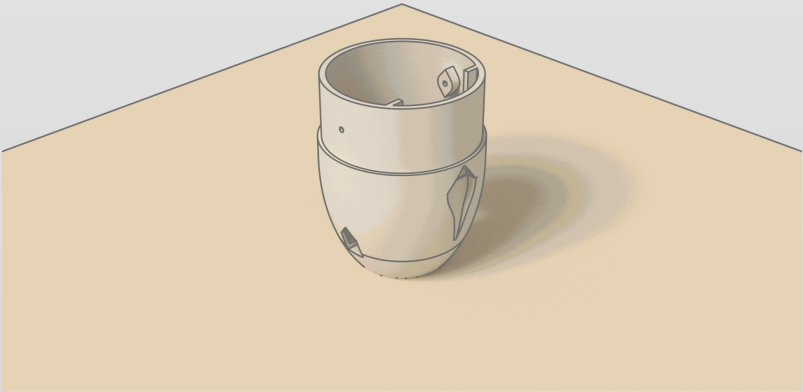
PARTS LIST

PART	MATERIAL	MASS
N0M NOSE (MICRO FPV)	PETG	19.7 g
N0P NOSE (PI CAMERA)	PETG	17.6 g
N0O NOSE (DJI O4)	PETG	25.4 g
F3 TAIL MODULE	ASA	15.4 g
HF FRONT HATCH	PETG	4.2 g
FN FIN	ASA	6.7 g
VTX SHELF (N0M)	PETG	1.0 g
HG GPS CARRIER	PETG	1.2 g
NB1 WING-NUT BAR FRONT	PETG-CF	1.3 g
NB2 WING-NUT BAR REAR	PETG-CF	1.2 g
HL1 HATCH LIP FRONT	PETG-CF	0.9 g
HL2 HATCH LIP REAR	PETG-CF	0.9 g

PART	MATERIAL	MASS
WR WING HALF RIGHT	PLA-CF	47.2 g
WL WING HALF LEFT	PLA-CF	47.3 g
AR AILERON RIGHT	PETG-CF	4.0 g
AL AILERON LEFT	PETG-CF	3.8 g
SR TAIL PANEL RIGHT	PETG-CF	5.2 g
SL TAIL PANEL LEFT	PETG-CF	5.2 g
M1 MOTOR PLATE	PETG-CF	2.3 g
SP SPINNER	PETG-CF	1.6 g
TR EQUIPMENT TRAY	PETG-CF	24.1 g
TC TAIL COUPLER	PETG-CF	10.4 g
CW COUNTERSUNK WASHER (PRINT 14)	PETG-CF	0.1 g
WF WING-WINDOW FILLER	PETG-CF	1.3 g

One aircraft with the N0M nose: 269 g printed. Only one camera nose is needed; the others are options.

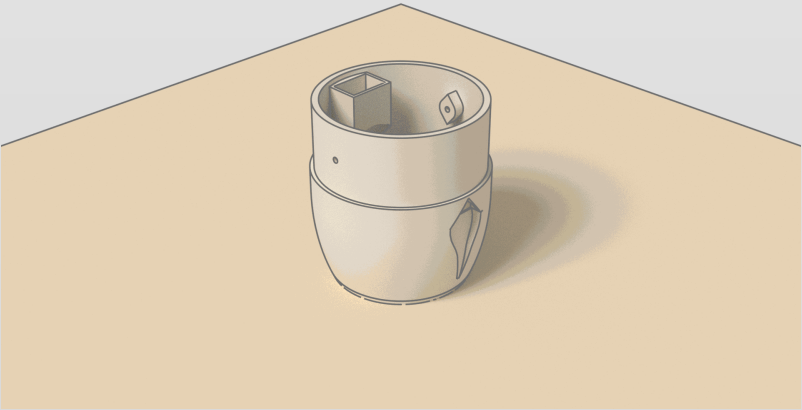
PARTS ORIENTATION



N0M NOSE (MICRO FPV)

FACE DOWN, SPIGOT UP

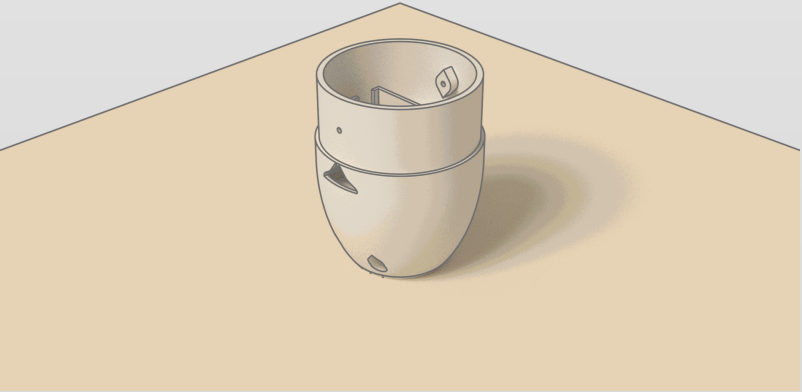
2 WALLS · 30 % INFILL · PETG



N0P NOSE (PI CAMERA)

FACE DOWN, SPIGOT UP

2 WALLS · 30 % INFILL · PETG

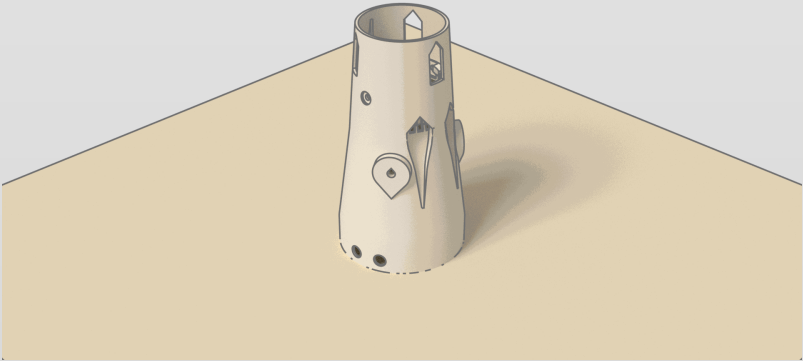


N0O NOSE (DJI O4)

FACE DOWN

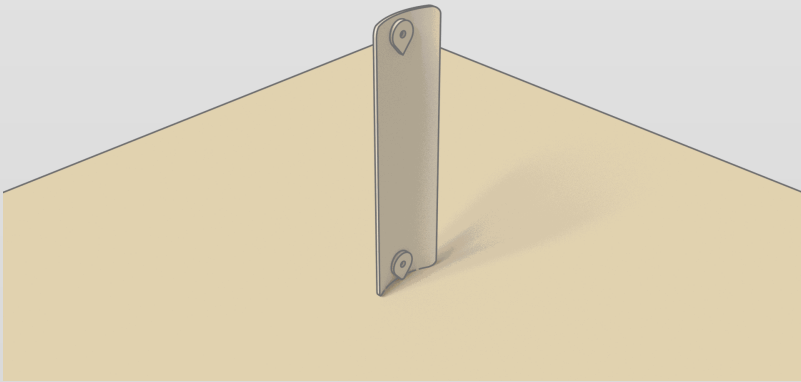
2 WALLS · 30 % INFILL · PETG

PARTS ORIENTATION



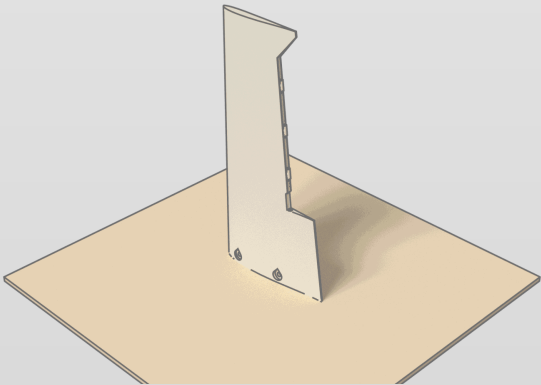
F3 TAIL MODULE

FRONT END (SOCKET) DOWN · 5 mm BRIM
2 WALLS · 30 % INFILL · ASA



HF FRONT HATCH

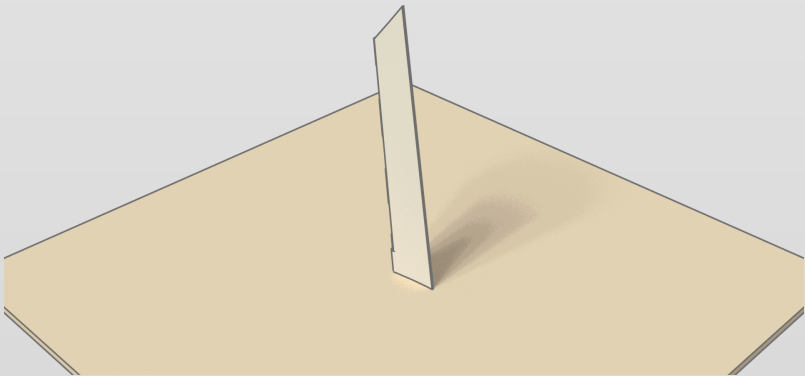
STANDING, FRONT END DOWN
2 WALLS · 30 % INFILL · PETG



WR WING HALF R / L

ON THE ROOT RIB
2 WALLS · 8 % INFILL · PLA-CF

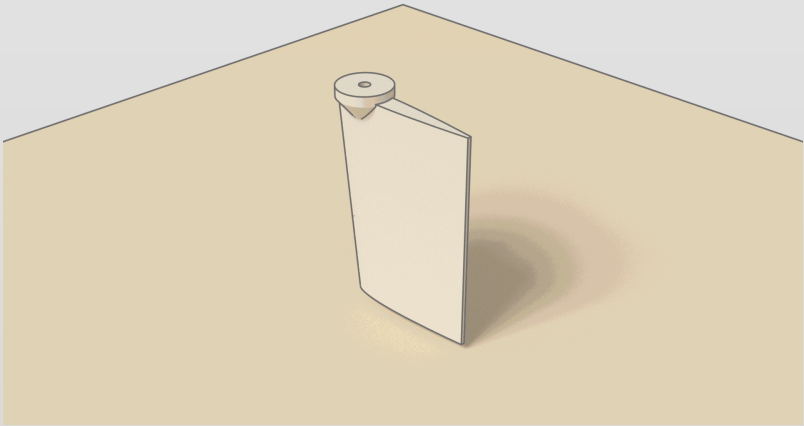
PARTS ORIENTATION



AR AILERON R / L

ON THE INBOARD END

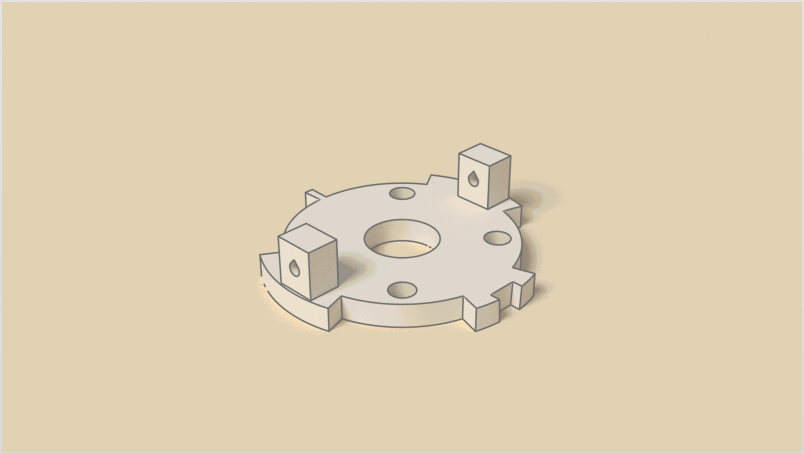
1 WALLS · 5 % INFILL · PETG-CF



SR TAIL PANEL R / L

TIP DOWN, HUB UP

2 WALLS · 10 % INFILL · PETG-CF

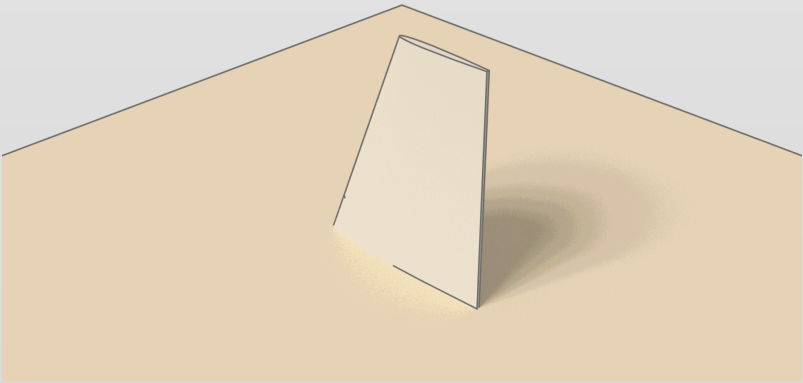


M1 MOTOR PLATE

FLAT

4 WALLS · 40 % INFILL · PETG-CF

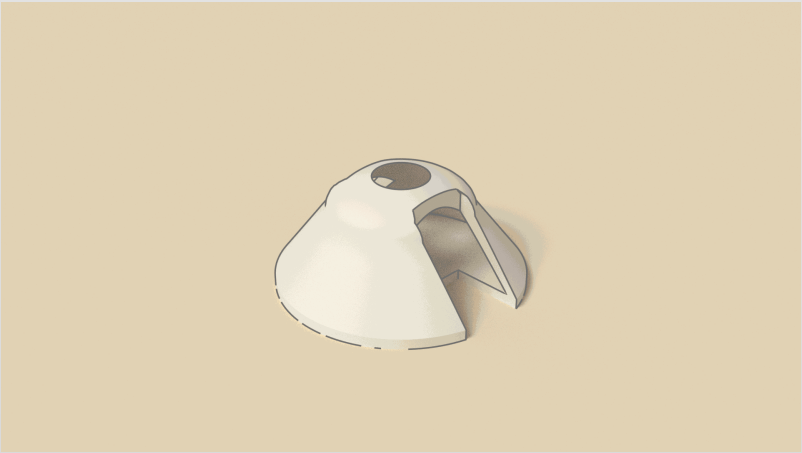
PARTS ORIENTATION



FN FIN

ON THE ROOT TAB

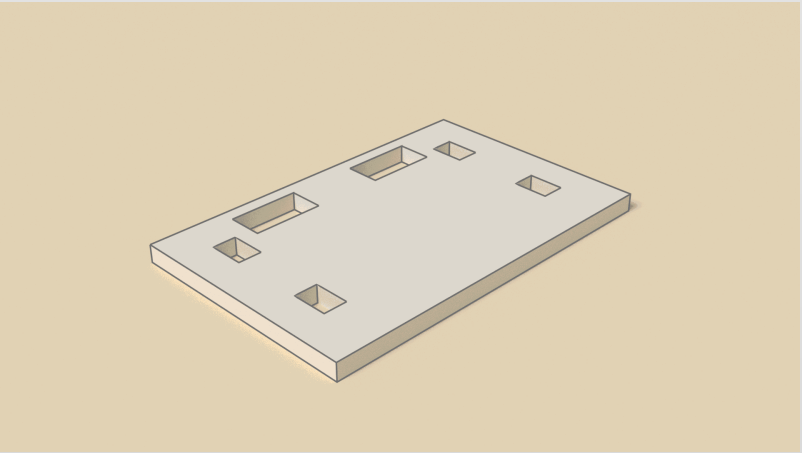
2 WALLS · 30 % INFILL · ASA



SP SPINNER

BACKPLATE DOWN

4 WALLS · 40 % INFILL · PETG-CF

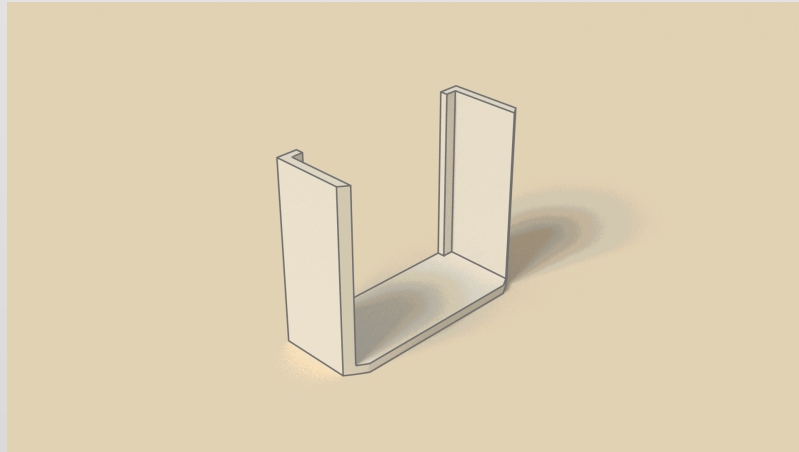


VTX SHELF (NOM)

FLAT

2 WALLS · 30 % INFILL · PETG

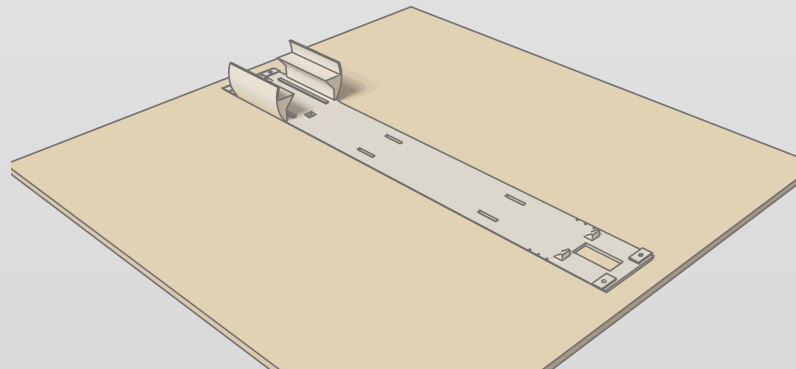
PARTS ORIENTATION



HG GPS CARRIER

ON ITS REAR WALL

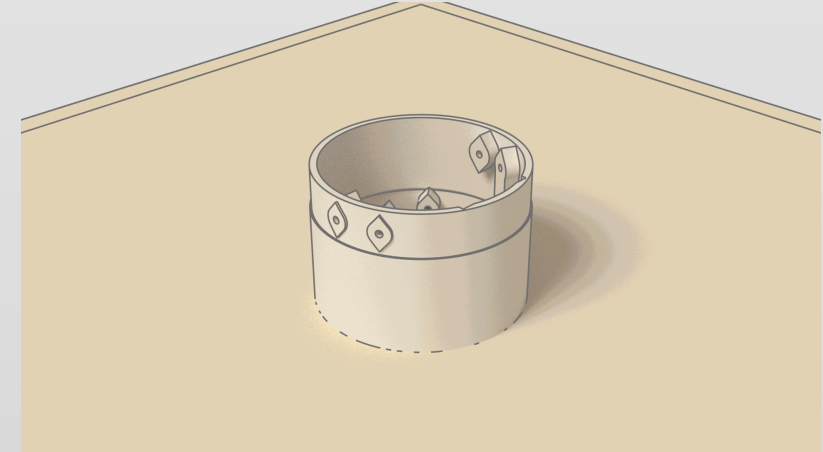
2 WALLS · 30 % INFILL · PETG



TR EQUIPMENT TRAY

UPRIGHT ON ITS FLAT UNDERSIDE

2 WALLS · 30 % INFILL · PETG-CF

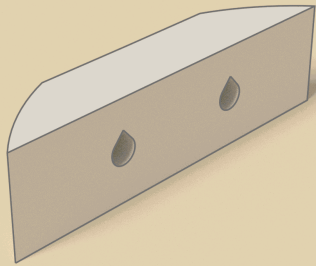


TC TAIL COUPLER

ON ITS SPIGOT END

2 WALLS · 30 % INFILL · PETG-CF

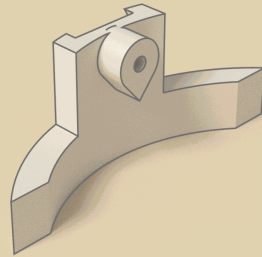
PARTS ORIENTATION



NB1 WING-NUT BAR FRONT

ON ITS END FACE

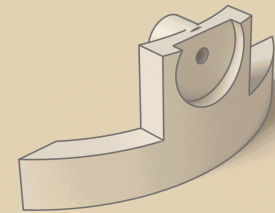
2 WALLS · 30 % INFILL · PETG-CF



HL1 HATCH LIP FRONT

BONDING FACE DOWN

2 WALLS · 30 % INFILL · PETG-CF

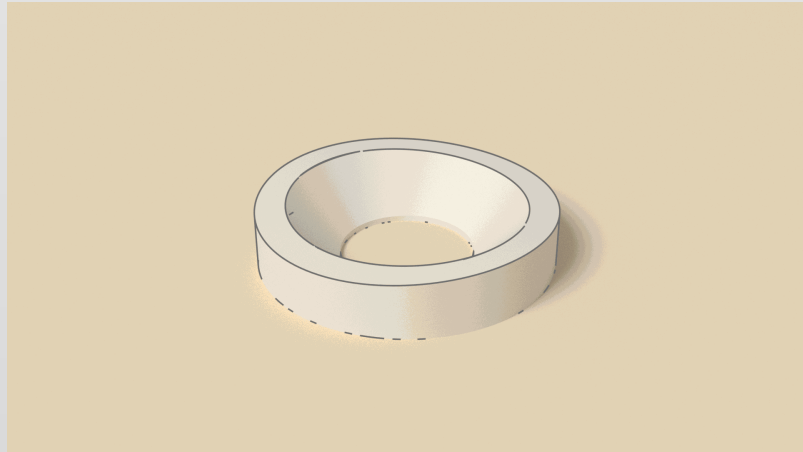


HL2 HATCH LIP REAR

BONDING FACE DOWN

2 WALLS · 30 % INFILL · PETG-CF

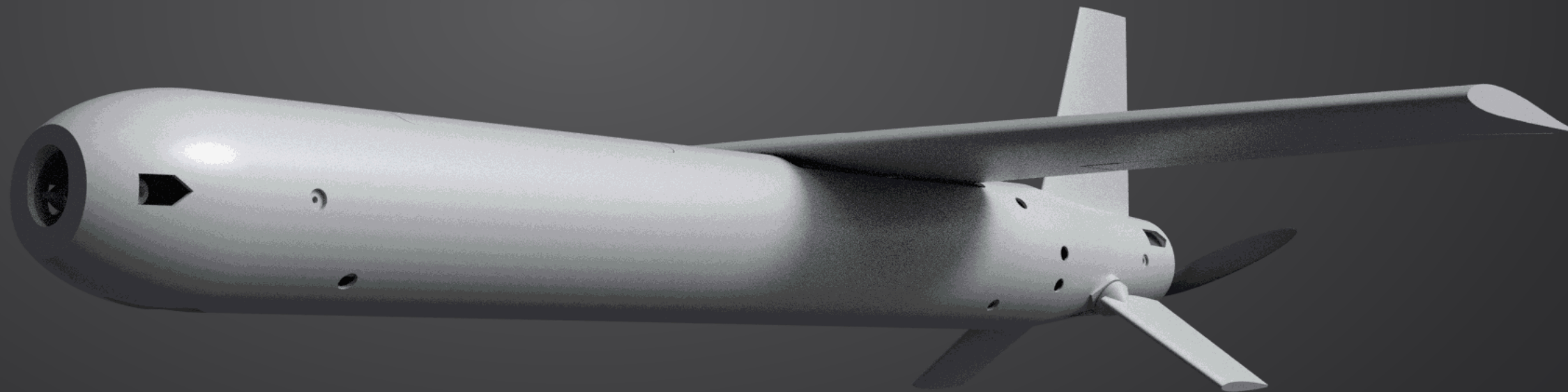
PARTS ORIENTATION



CW COUNTERSUNK WASHER (PRINT 14)

FLAT FACE DOWN

2 WALLS · 30 % INFILL · PETG-CF



ASSEMBLY GUIDE

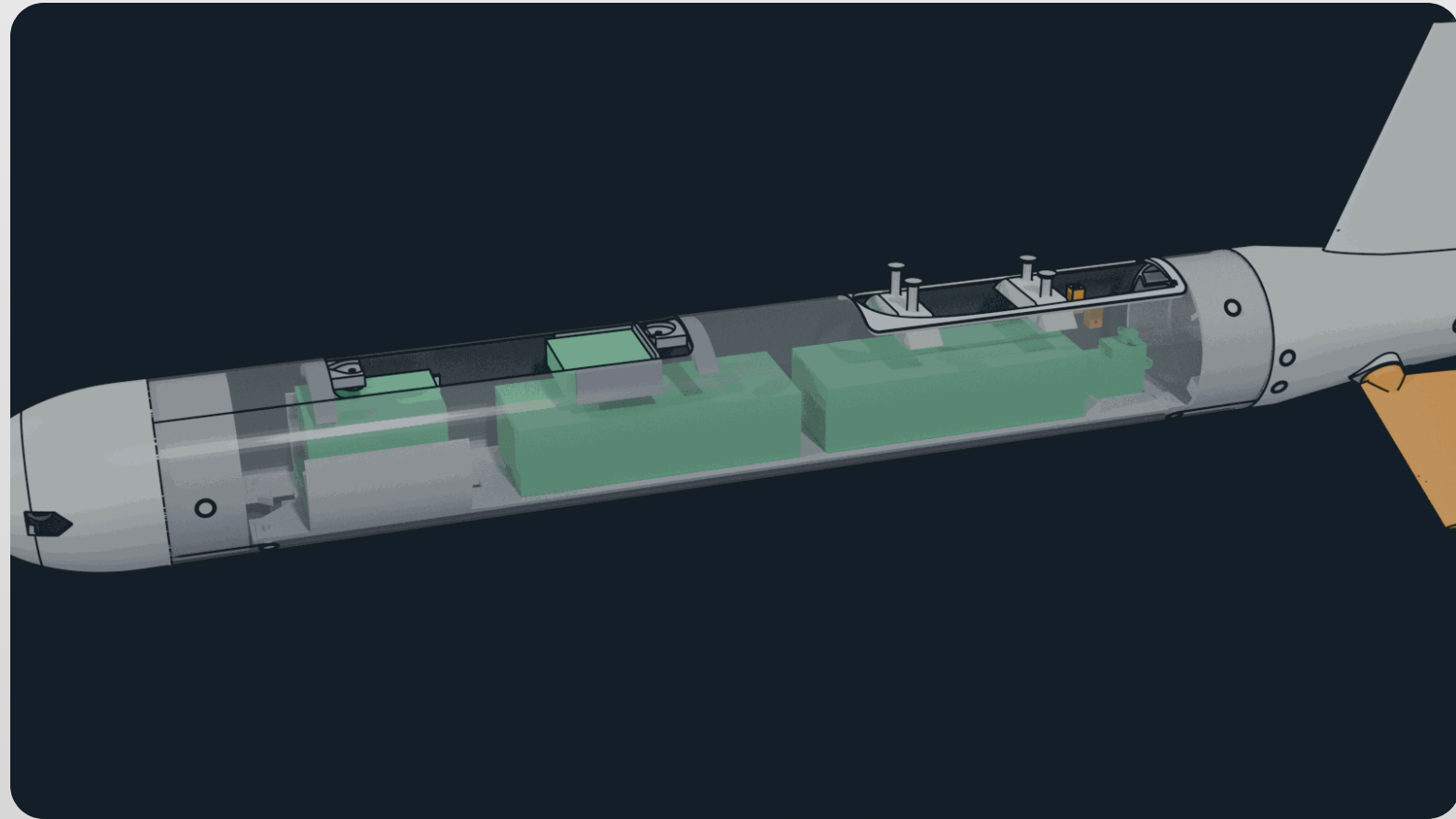
CARBON TUBE



Cut the 50 × 48 mm carbon tube to 320 mm. Print the 1:1 cutting template (two A4 pages, 100 % scale: check its 100 mm bar), cut the strip out, wrap it round the tube with its ends on the tube ends and the TOP line along the top, tape it. Cut every window inside the red lines with a diamond disc and file to the line; drill the 2.2 mm holes. FFP3 mask, eye protection, vacuum at the disc or cut wet. Seal every cut edge and hole with thin CA: carbon conducts, a bare edge can cut a wire's insulation and short it.



DIAMOND DISC



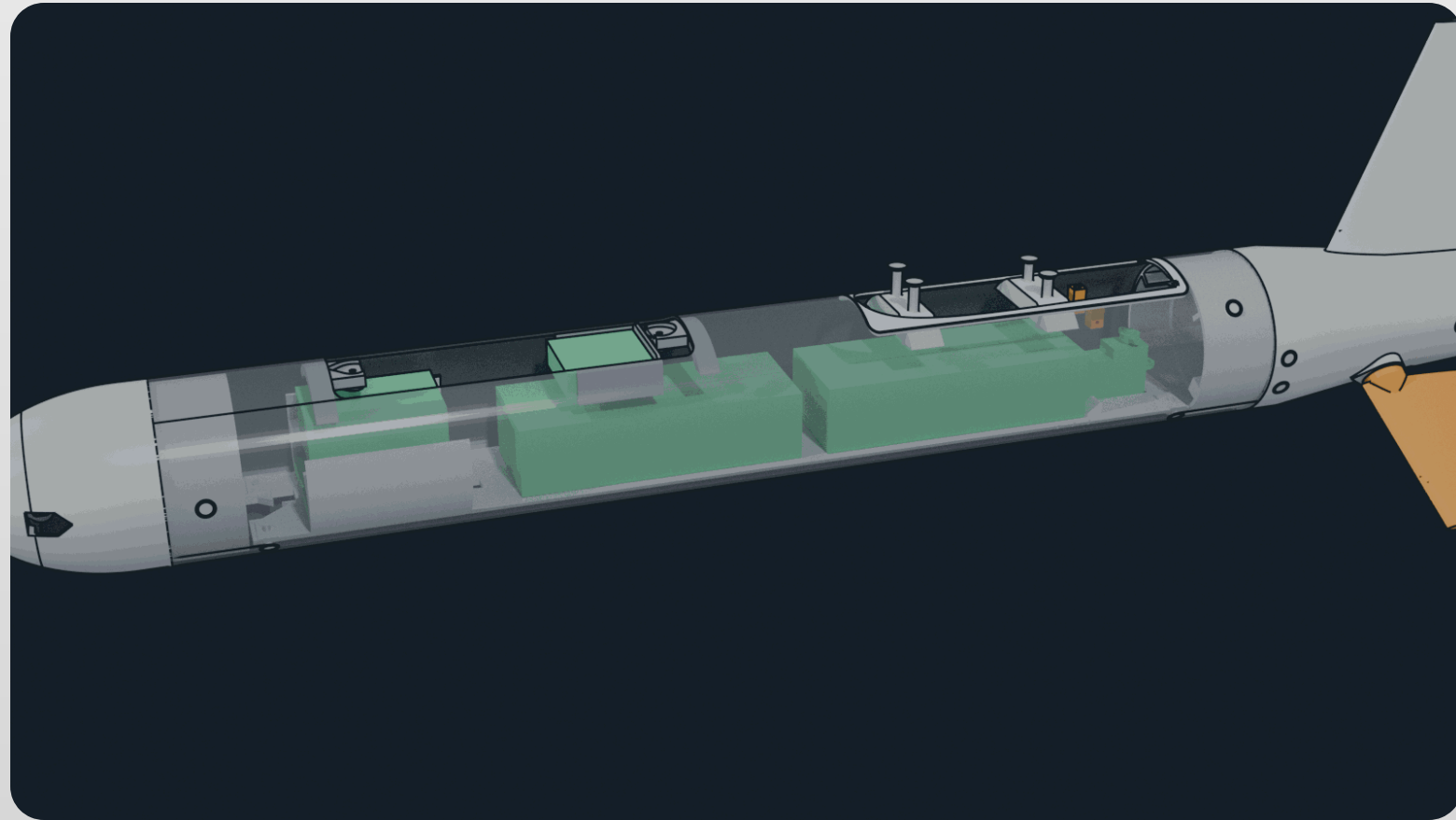
CARBON TUBE



Scuff the bonding areas inside the tube (120 grit) and clean them with alcohol. With structural epoxy bond: the tail coupler TC into the aft end, its sleeve over the whole ring behind the wing window, and fix it with 2 × countersunk M2 × 6 in washers at the sides (they hold it aligned while the epoxy cures and back the bond up); the two wing-nut bars NB1/NB2 across the wing-bay window (in through the window, turned under its edges, pocket down); the hatch lips HL1/HL2 under the front hatch opening's ends (their screw bosses down, centred under the opening). Let it cure before the tray goes in.



EPOXY



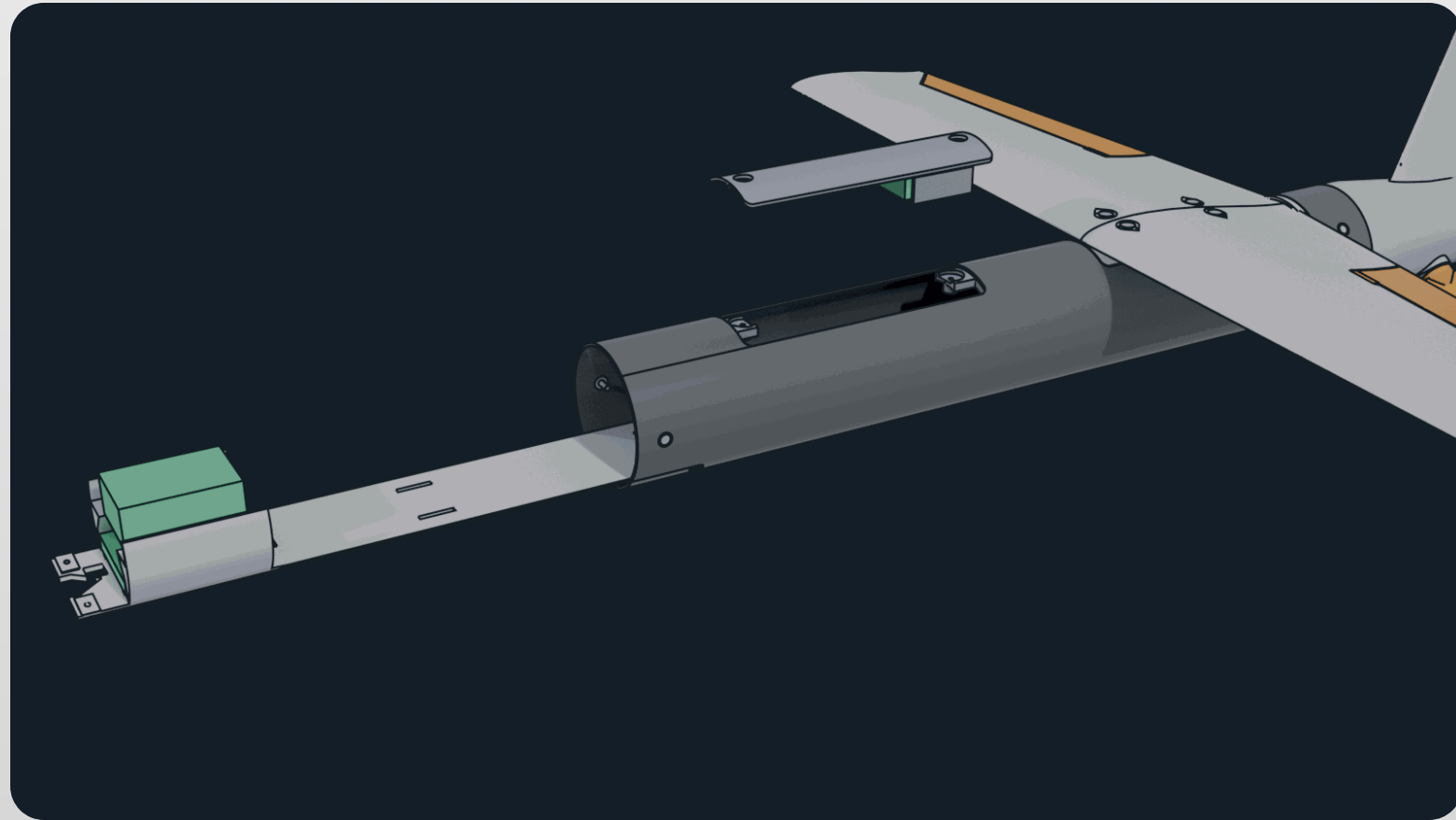
EQUIPMENT TRAY



Build the tray on the bench: the ESC under the FC ledges, the FC stack, receiver, aileron servo, the four packs and all wiring as on the next pages. Take the camera nose off, push the tray into the tube from the front until its screw holes line up, and fix it with 4 × countersunk M2 × 6 screws through the tube (do not overtighten: they thread into printed pillars). Every external screw (the tube, the tail joint, the motor plate) sits flush in a printed countersunk washer (CW) pressed into its 5 mm hole: print 14, push one into each hole before the screw. To service the equipment or inspect the packs: nose off, unhook the aileron pushrods, unplug the tail-module plugs, four screws out, slide the tray forward (the packs come out with it).



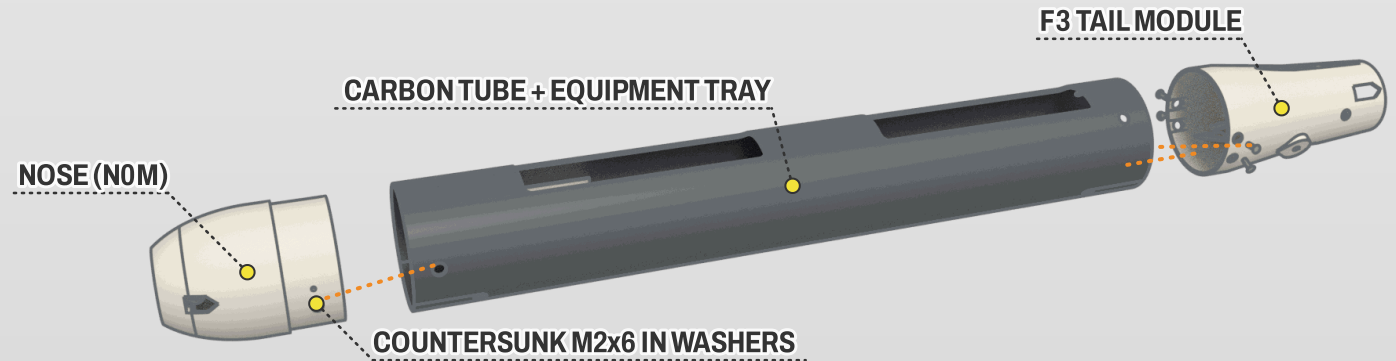
COUNTERSUNK M2 × 6



FUSELAGE ASSEMBLY



The camera nose plugs into the tube's front (its spigot is sized to the tube's bore) with 2 × countersunk M2 × 6, flush in their printed washers, into its bosses; the tail module slides onto the tail coupler's spigot (the four pads on the coupler slide in the grooves inside the module) and is held by 4 × countersunk M2 × 6 (two each side: at the side and 25° below it), each in a printed washer in its 5 mm hole, as on the tube. Nothing is glued: both modules come off for service. They must slide on by hand; if the bore is tight, sand the spigot, never force it.

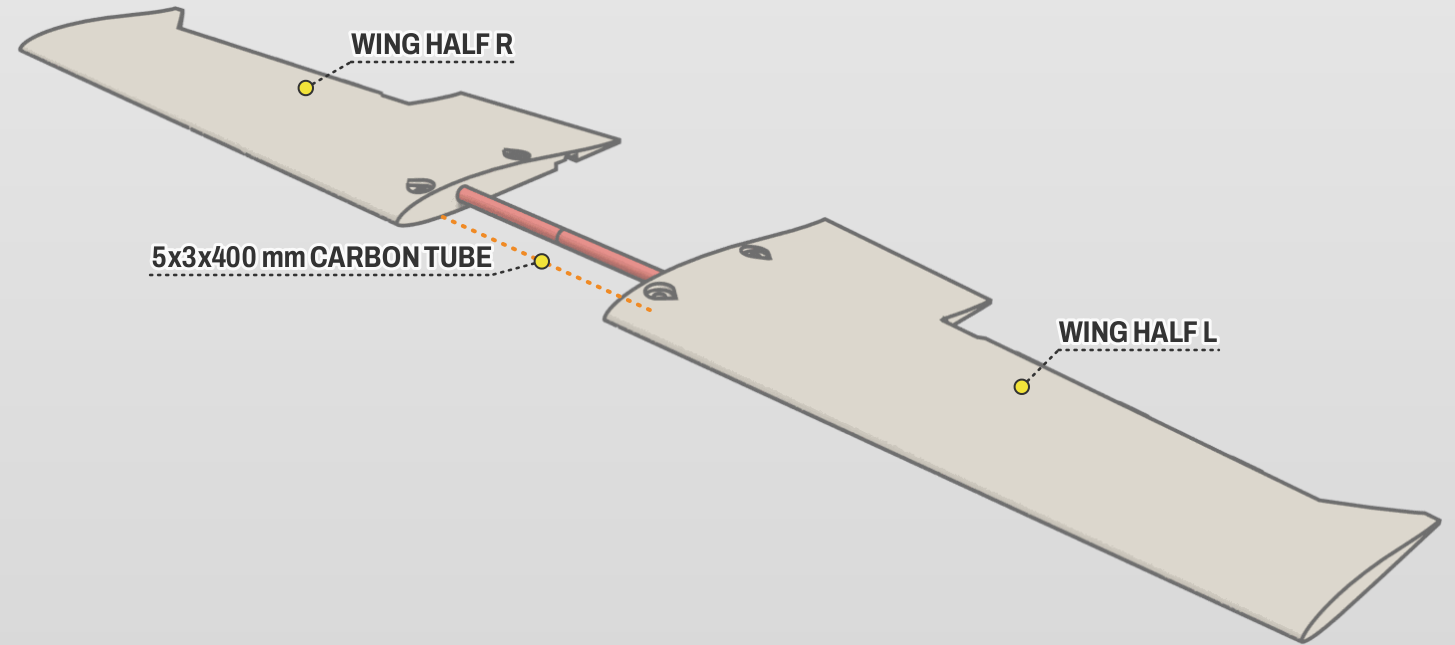


M2 × 6 SCREWS

WING ASSEMBLY



Cut one 400 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (200 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).

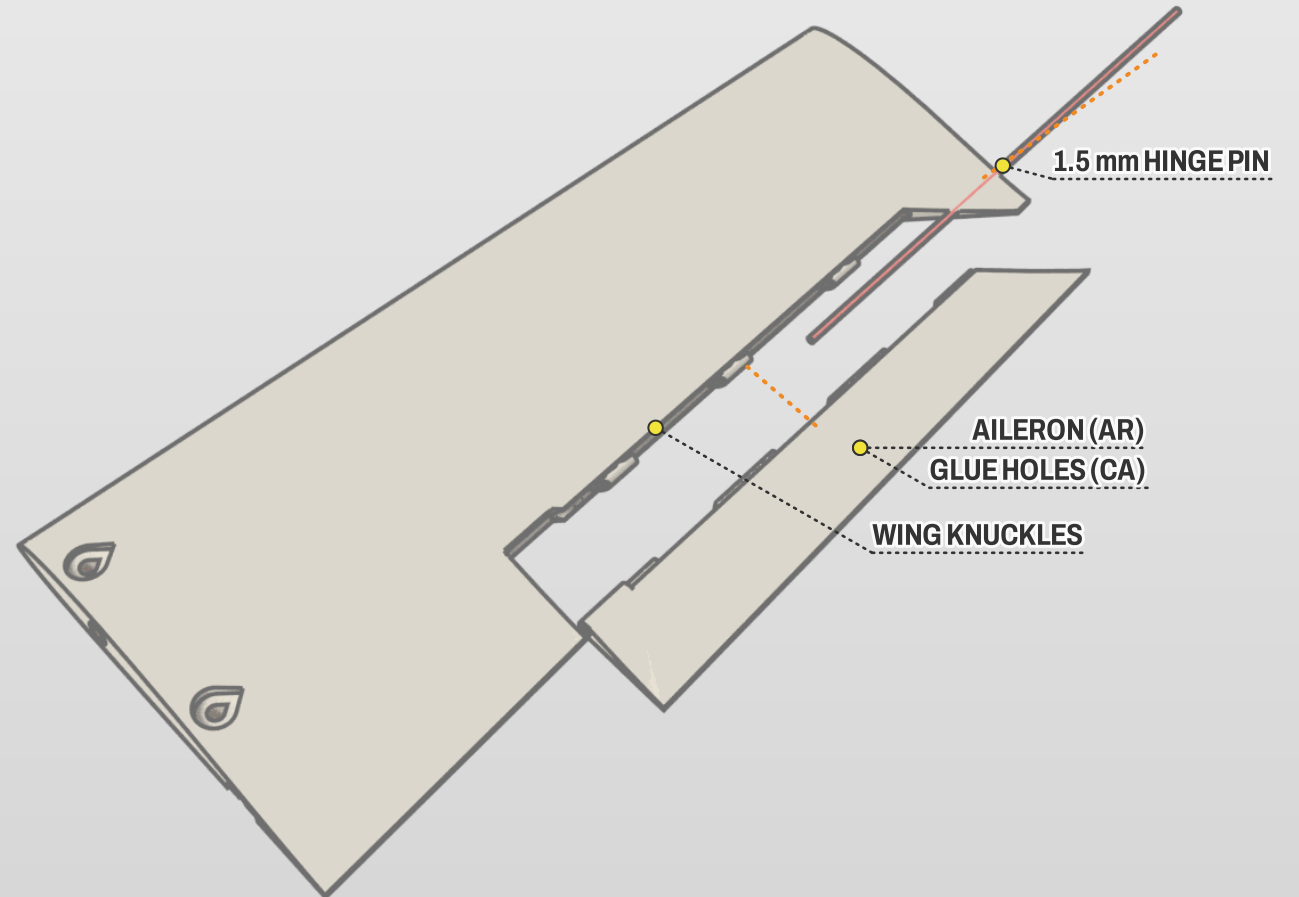


EPOXY

WING ASSEMBLY



Set each aileron in place: its rounded nose sits in the cove along the wing's hinge face and the wing's three knuckles enter the pockets in the nose. Push a 132 mm length of 1.5 mm steel rod in from the wingtip through the tip, the aileron and the knuckles. Wick thin CA into the aileron's four glue holes only, then plug the tip hole. The aileron must swing freely.

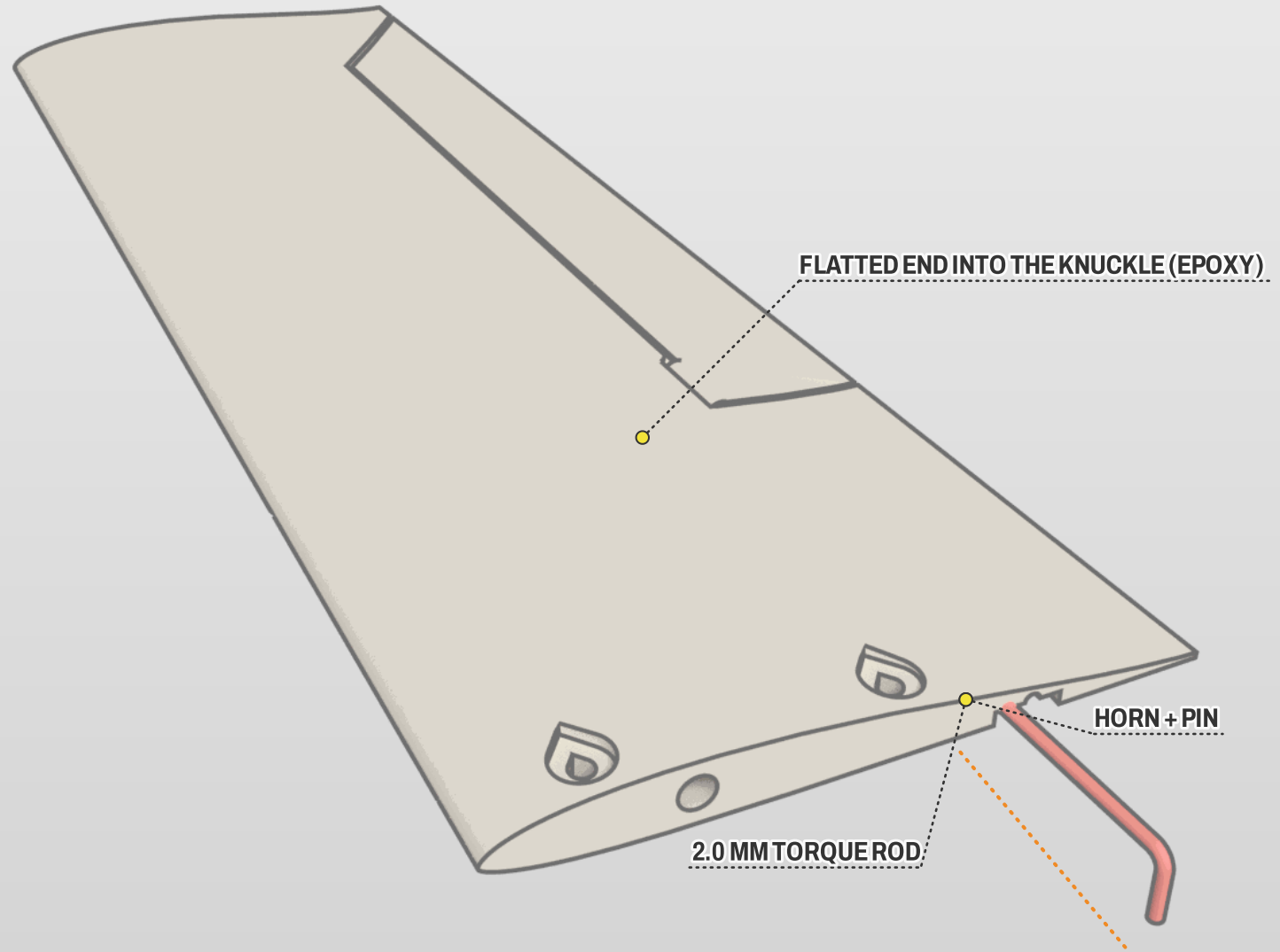


THIN CA

WING ASSEMBLY



Bend a 12 mm horn down at one end of each 2.0 mm rod with a 5.5 mm pin outboard at its tip, and file a 0.3 mm flat on the last 9 mm of the straight end, opposite the horn. Push it in from the root face until the flat enters the D-bore of the aileron's root knuckle; epoxy the end and hold the aileron 3° trailing edge up while it cures.

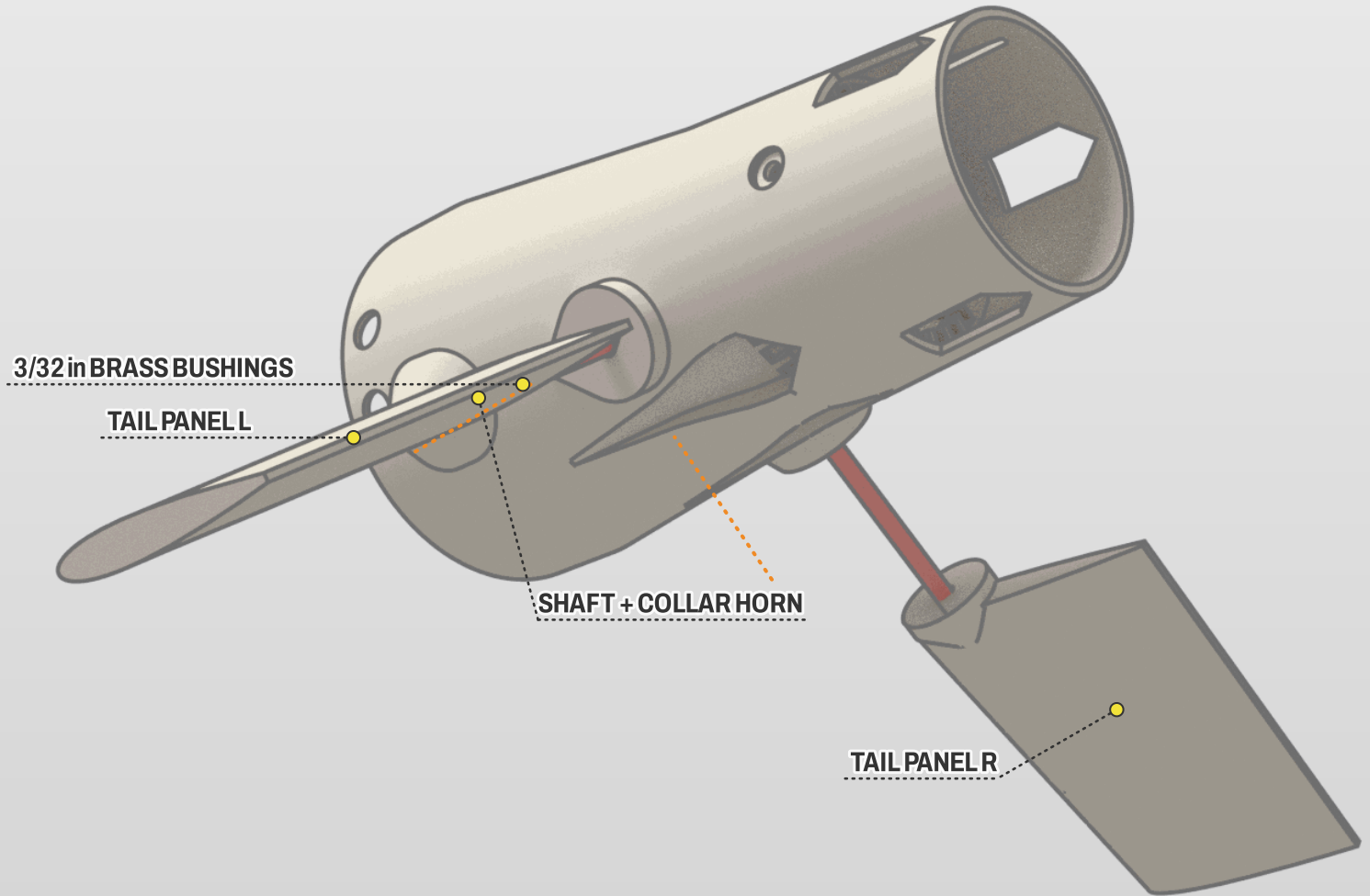


EPOXY

TAIL MODULE



Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. From the open front of F3, push each shaft through its bushing and fit the wheel-collar horn with its 8 mm brass plate standing along the panel's normal (not pointing forward), 0.5 mm end play, set screw on a filed flat.

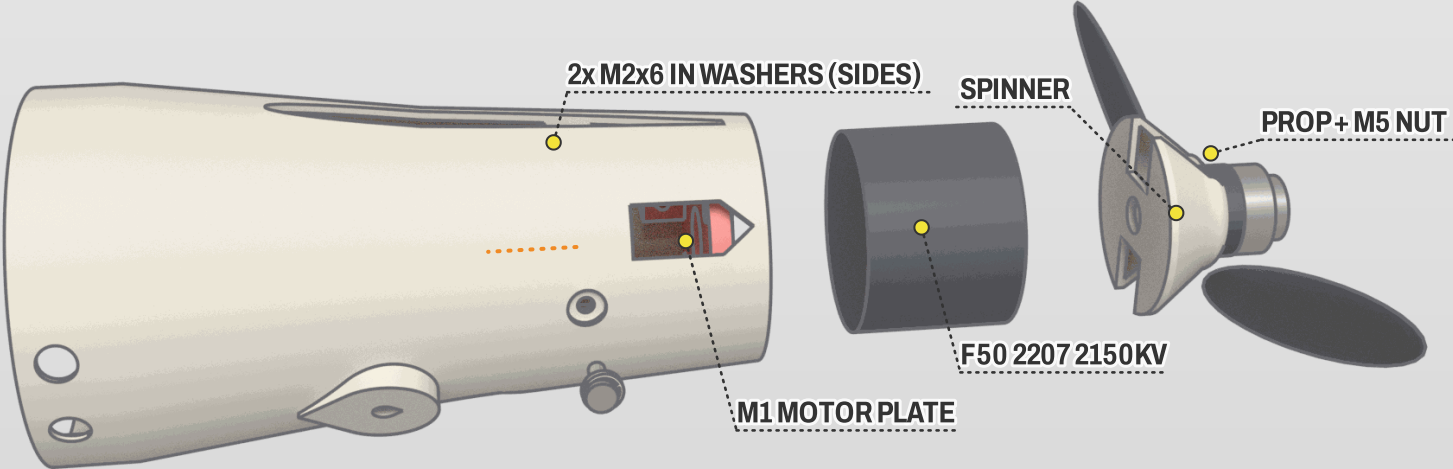


EPOXY

TAIL MODULE



Screw the motor to the M1 plate on the bench with its 4 × M3 (heads on the plate's front face). Slide the plate into the rear of F3 until it sits on the seat ring, its two side lugs in the ring's pockets and its edge notches on the cooling windows, then fix it with 2 × countersunk M2 × 6 through the tail module's sides into the lugs, each in a printed washer in its counterbore, flush (no glue: the plate comes out for a different motor). Fit the spinner backplate, the prop (Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre polyamide, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring) and the thin M5 nut.



M2 × 6 SCREWS

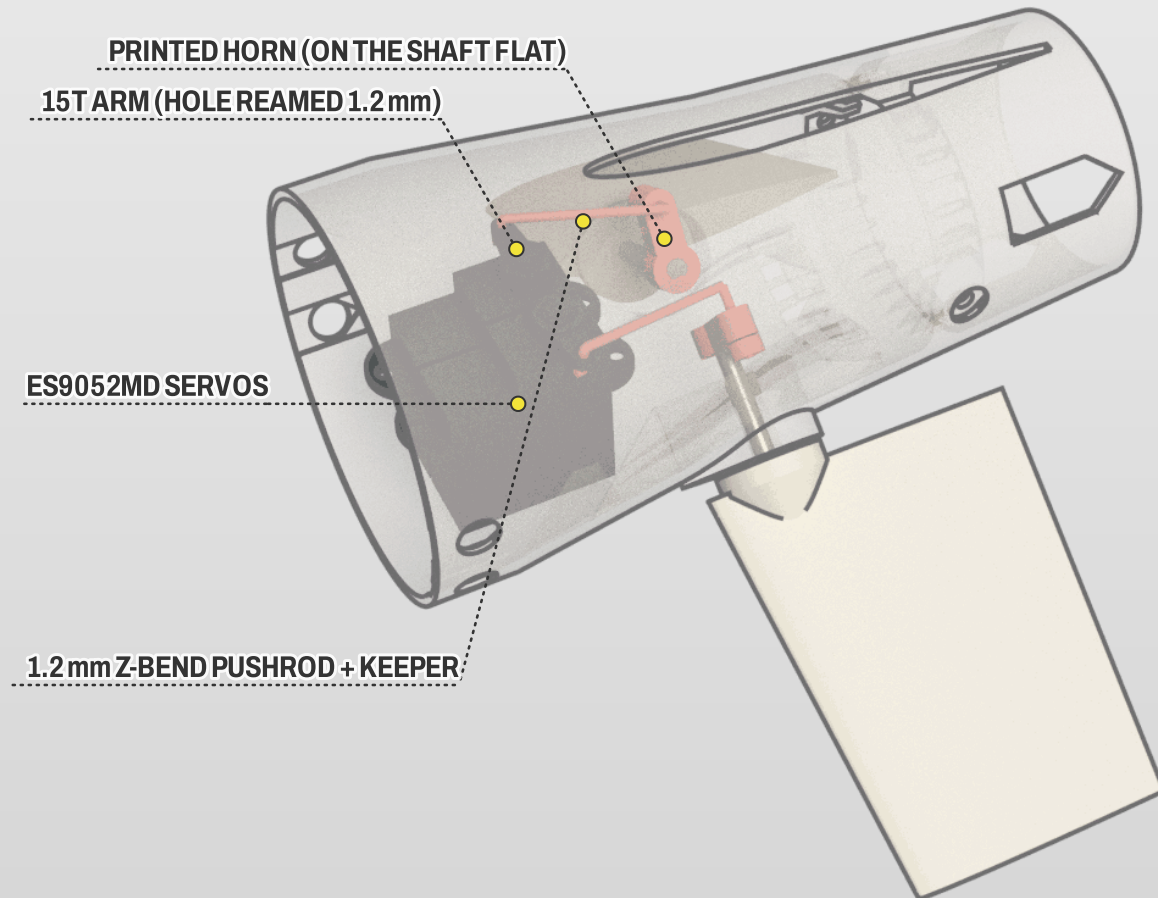
TAIL MODULE



Stand both ES9052MD servos through the cut-outs in the F3 floor on foam tape on the bottom skin, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; file a 0.3 mm flat on the inboard end of each 2.5 mm stabilator shaft and epoxy the printed horn (THR/THL) on it, arm along the panel's normal. Drill the servo arm's 6.7 mm hole 1.1 mm and ream it by hand with a 1.2 mm bit (and the horn's hole the same): the wire must turn with no play. 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend at the horn end pushed outboard through the horn's reamed hole, snap-on keeper on the protruding end (unclip it to take the tail module off). Free play at each panel trailing edge ≤ 0.2 mm. Check both panels move freely through $\pm 20^\circ$.



SERVO TAPE



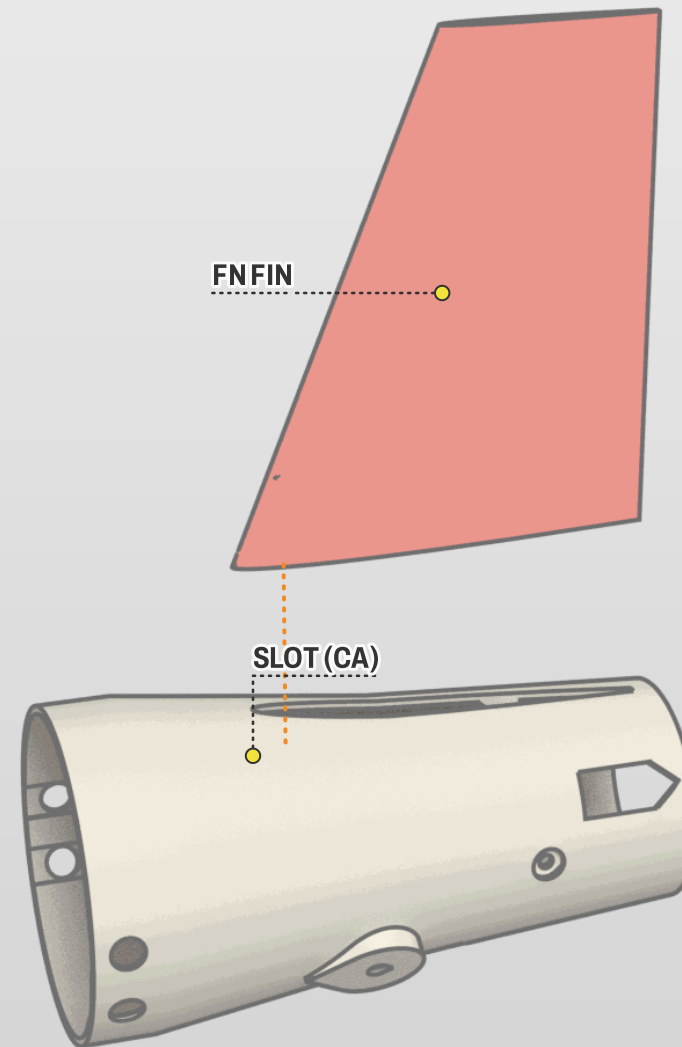
TAIL MODULE



Glue the fin into the top slot of F3 with CA, vertical and square to the wing.



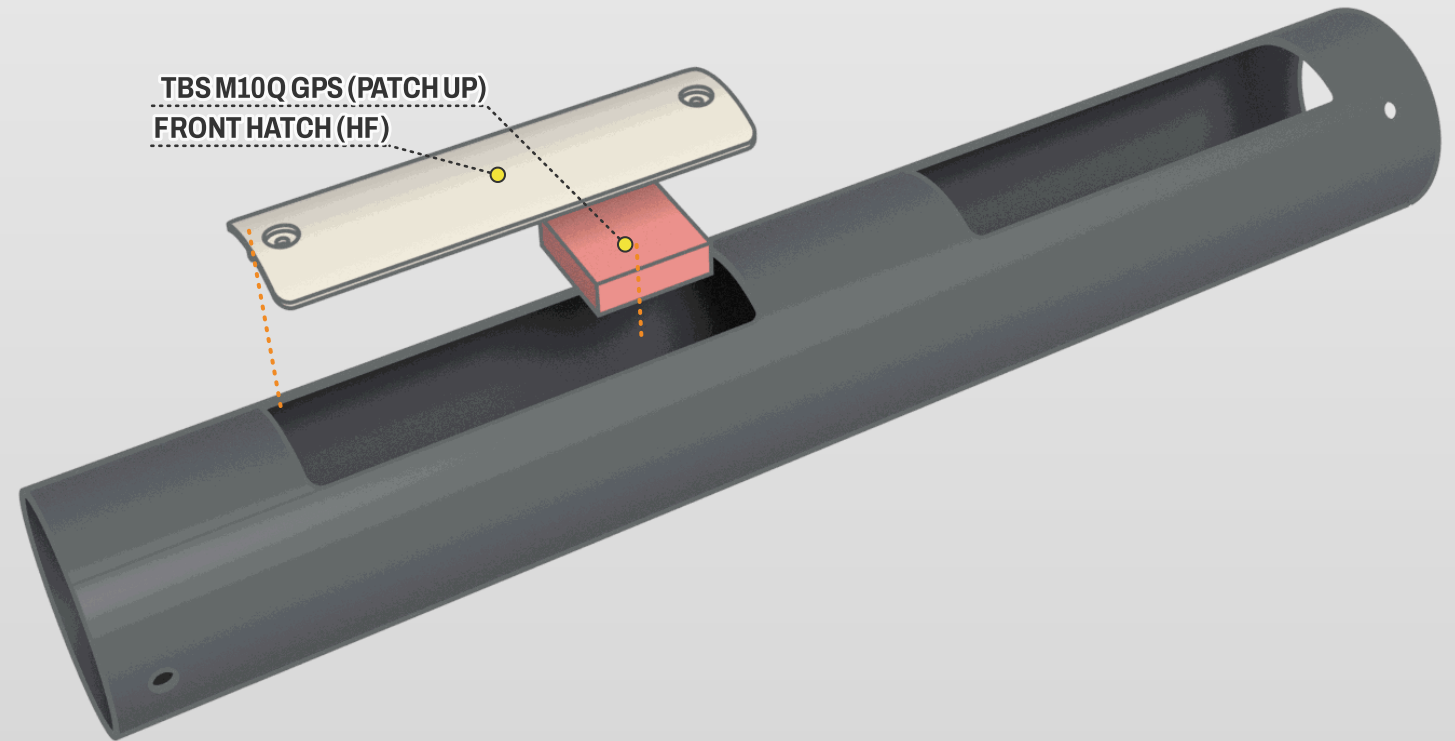
CA GLUE



FUSELAGE ASSEMBLY



Bond the printed GPS carrier (HG) under the rear of the PETG hatch cover (the GPS's radio window: never paint it with metallic paint). Put foam tape on the carrier's two side lips and slide the TBS M10Q GPS in from the front, patch up, onto the lips against the rear wall: it is carried from below, with nothing but air and the cover between its patch and the sky. The cover is held by two flush countersunk M2 × 6 screws in steel cup washers, one at each end, into the bosses under the bonded lips (no magnets: suction and the GPS's weight would lift them at speed). Leave a slack loop in the GPS lead so the cover lifts off. With the cover off you reach the charge / arming port (XT30 and 6S balance plug), the FC's USB-C socket parked beside them (setup, parameters, logs) and the FC itself.

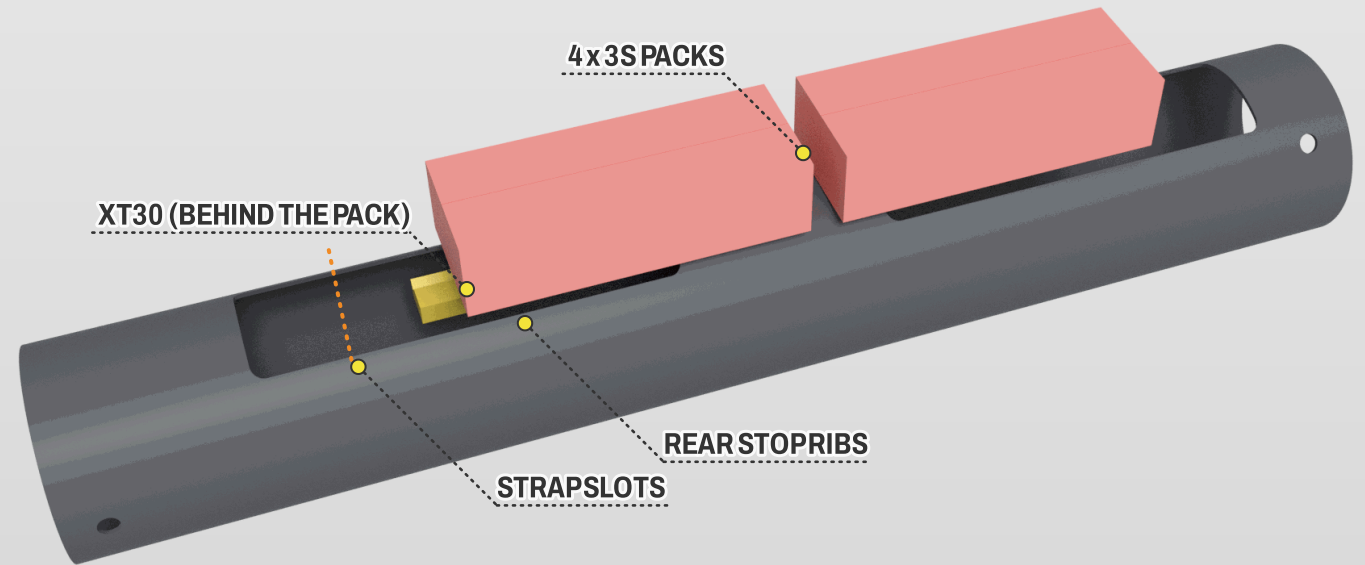


FOAM TAPE

FUSELAGE ASSEMBLY



The four packs stay in the aircraft: on the bench, lay them on the tray's rear floor as two side-by-side pairs (on their sides), the front pair's leads out of its rear end and the rear pair's out of its front end into the 8 mm gap. Set the rear pair's rear edge at the tick the balance table gives (ticks every 5 mm on the floor edges), foam tape under each pack and one strap round each pair through the slots. Plug each pair into its XT30 series adapter and 2 x 3S balance adapter (measure 6S, about 25 V full, and the polarity), then both pairs into the parallel adapter and balance board, only with the pairs within 0.05 V per cell; tape the adapters as one block on the rear pair. Lead the main XT30 and the 6S balance plug forward to the front pair, under the hatch: that is the charge / arming port.

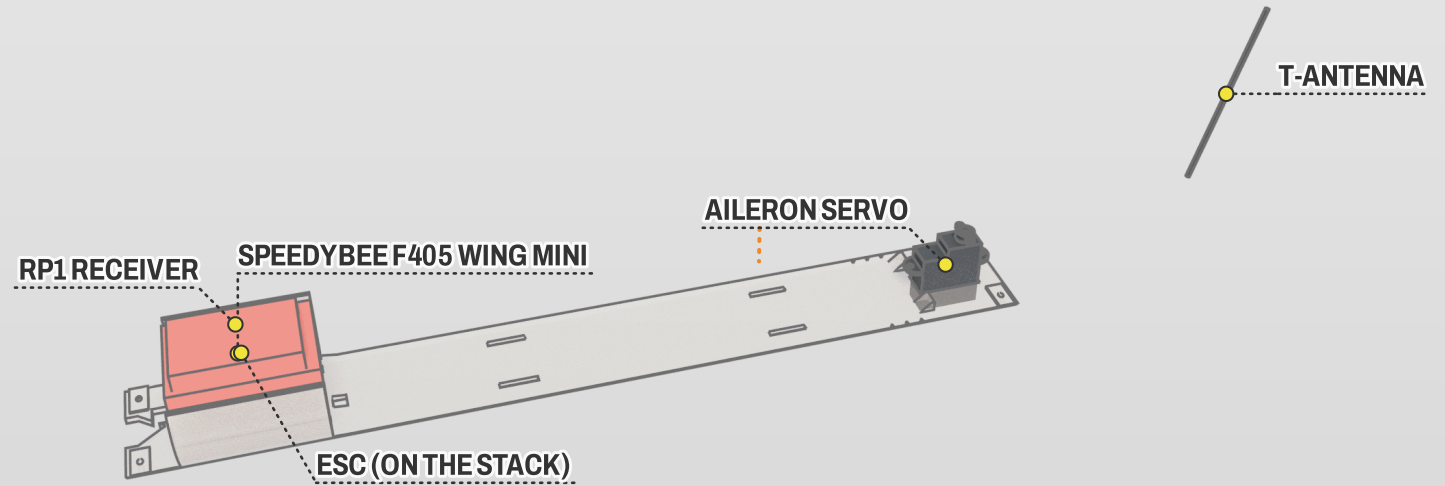


VELCRO STRAP

EQUIPMENT



Front bay of the tray: slide the HDZero Halo ESC in from behind under the two FC ledges, onto its rails (foam tape), motor on M1. SpeedyBee F405 WING MINI stack on the ledges above it on foam tape and a strap, arrow forward, servo BEC jumper at 6 V; the RP1 V2 receiver on edge on the right ledge (foam tape), its T-antenna straightened into the fin's dipole. Aileron servo through the tray's floor at the rear, output aft.

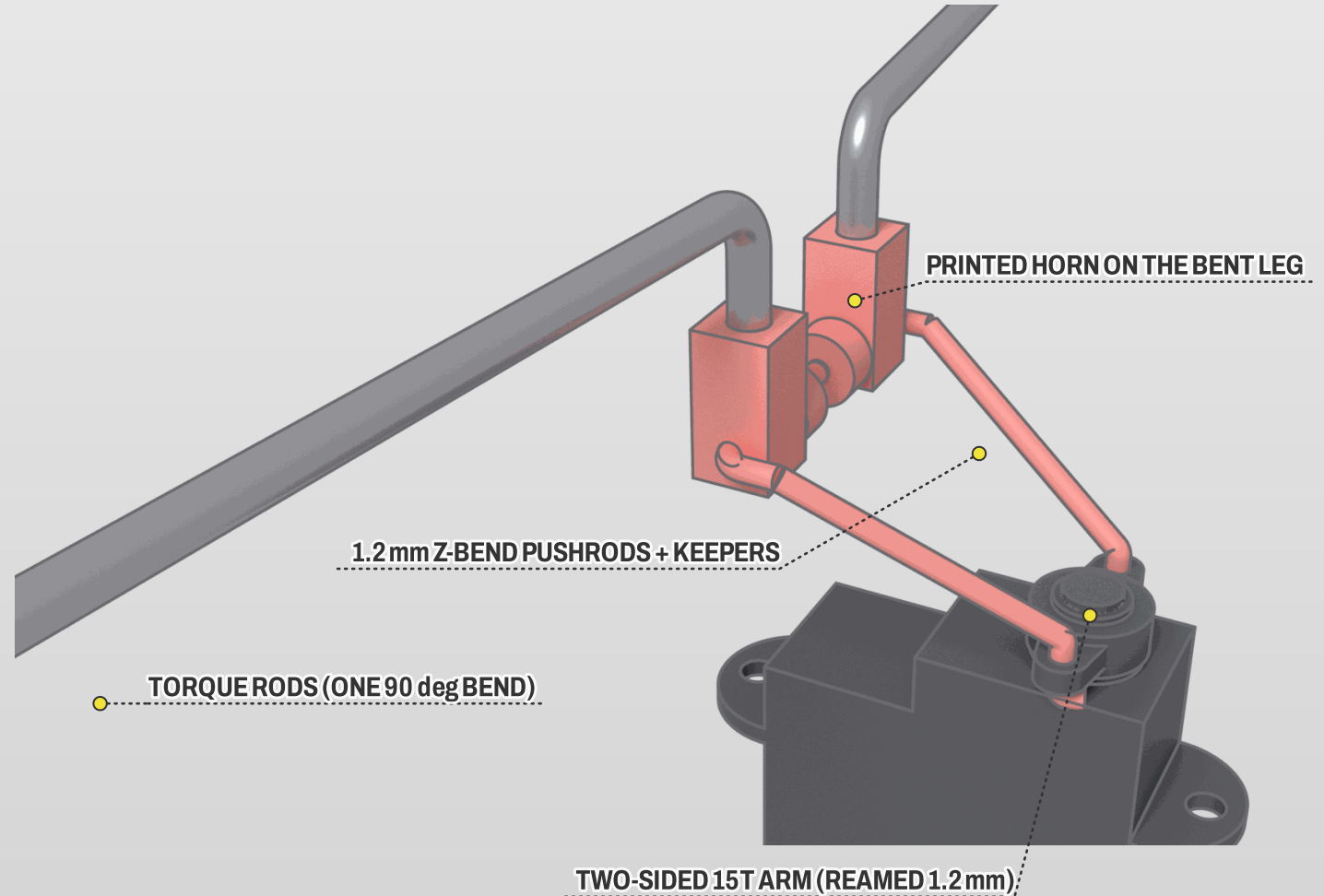


FOAM TAPE

EQUIPMENT



ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque rod has one 90° bend (the horn leg, 6 mm from the centreline); epoxy the printed horn (AHR/AHL) on the leg, 4 mm deep. Drill the arm's 4.3 mm holes 1.1 mm and ream them by hand with a 1.2 mm bit (and both horns' holes): the wire must turn with no play. Each horn connects to the arm with a 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend pushed inboard through the horn's reamed hole, snap-on keeper on the protruding end. Check: with the servo powered, each aileron trailing edge moves ≤ 0.2 mm (free play lowers aileron flutter to about 33 m/s). Set both ailerons 3° trailing edge up at servo neutral. No play at either end.

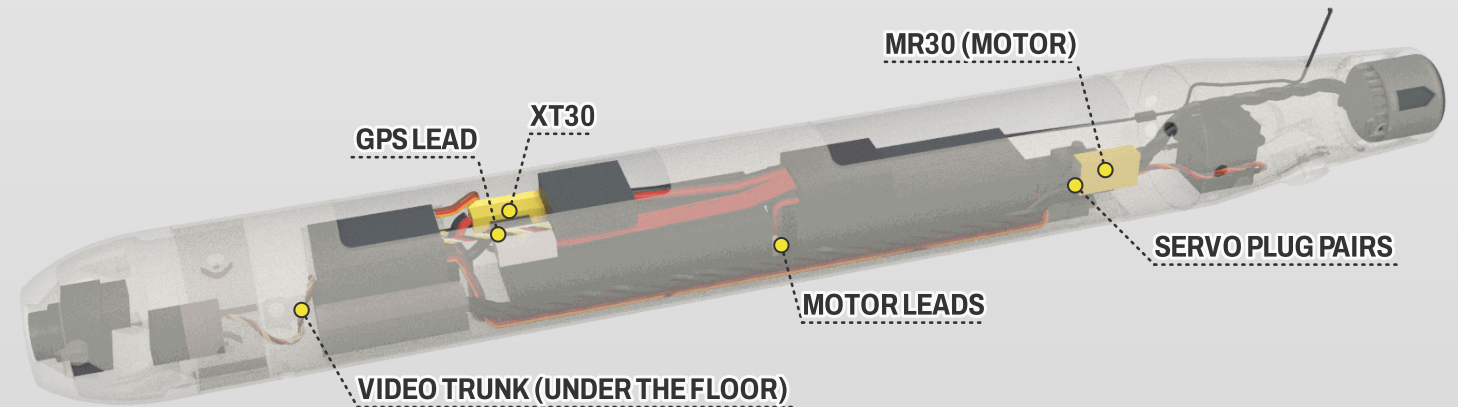


Z-BENDS

EQUIPMENT



Behind the FC: the battery lead from the port's XT30 down to the power board's BAT pads; the 14 AWG ESC feed from its ESC pads down onto the ESC's rear; ESC signal (M1, GND, TLM) up to S1; GPS and USB leads on the left. Motor phases from the ESC's rear down behind it and along the lower left wall past the packs to the MR30 beside the aileron servo; the left tail servo lead along the left wall below them; the right tail servo and aileron leads along the right wall (the aileron lead under the tray floor, up through the hole behind the ESC); the ELRS coax along the right side of the packs to the u.FL joint and into the fin. Cut lengths and lanes: Electrical Integration manual. Unplug the MR30, the u.FL joint and the servo plugs before pulling the tail module.



HEAT SHRINK

CAMERA NOSE



Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 8 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the shelf in from the rear onto its two rails and glue it; video antenna in the two clips on the right wall. Plug the one lead at the nose joint. Dome noses: glue the printed dome ring 3 mm into the front of the tube, then fit the clear dome over its lip with canopy glue (no CA).



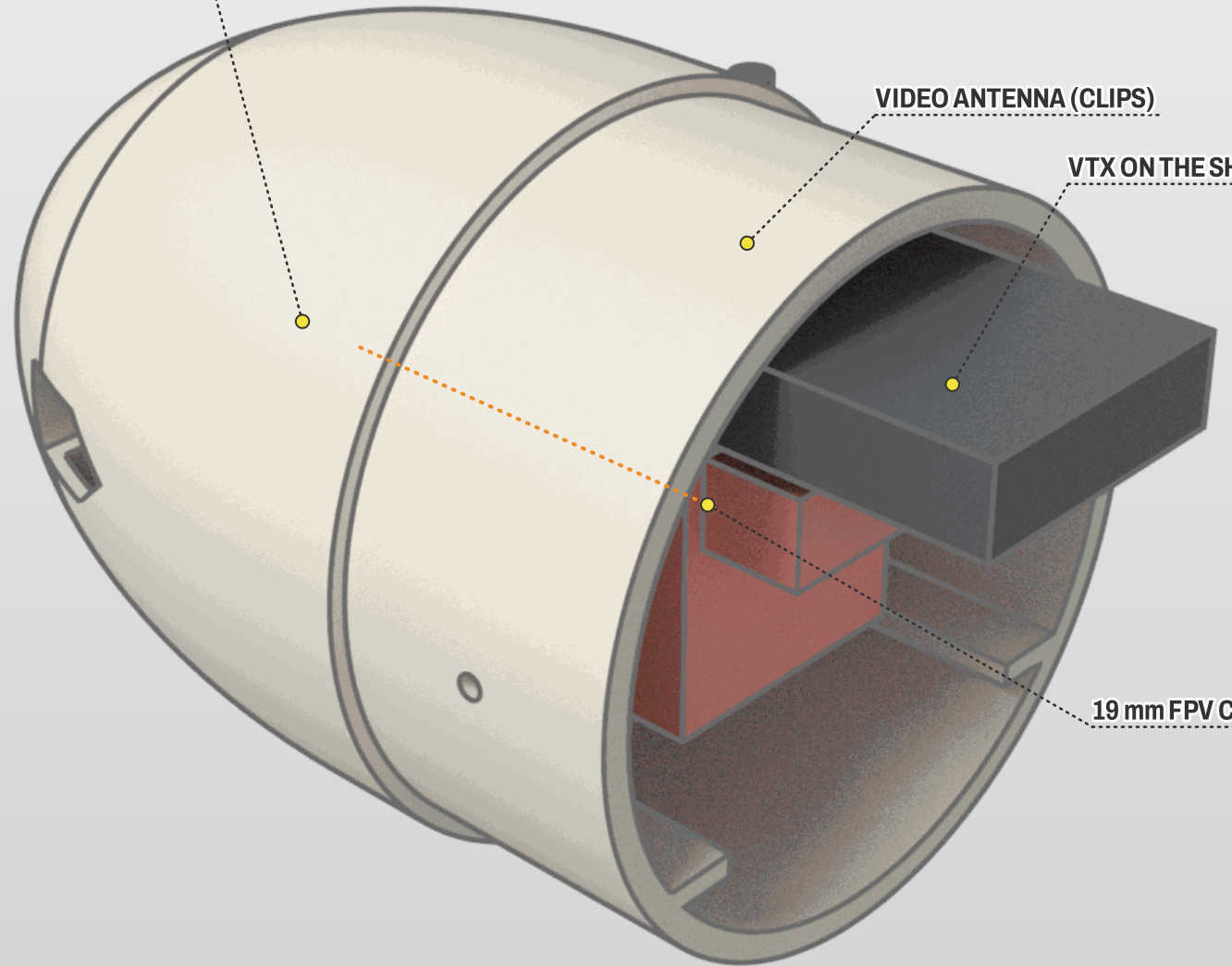
FOAM TAPE

2x M2x10 + WASHERS

VIDEO ANTENNA (CLIPS)

VTX ON THE SHELF

19 mm FPV CAMERA



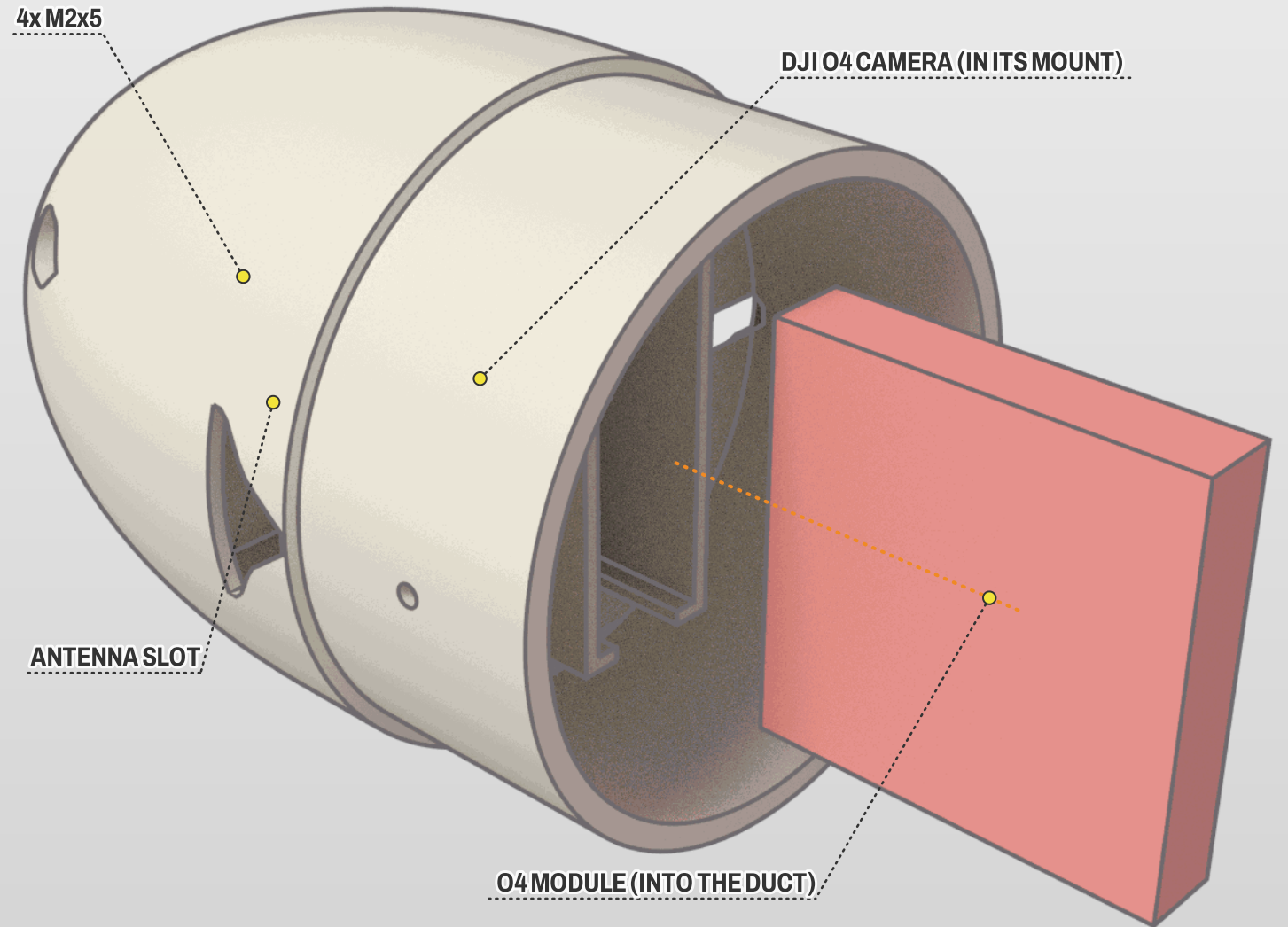
CAMERA NOSE



DJI O4 option: clamp the camera in its mount with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount). Plug the coax and 3-in-1 lead into the module and slide it edge-first into the duct until it stops; foam strip in the top groove, tape over the rear edge, never over its faces.



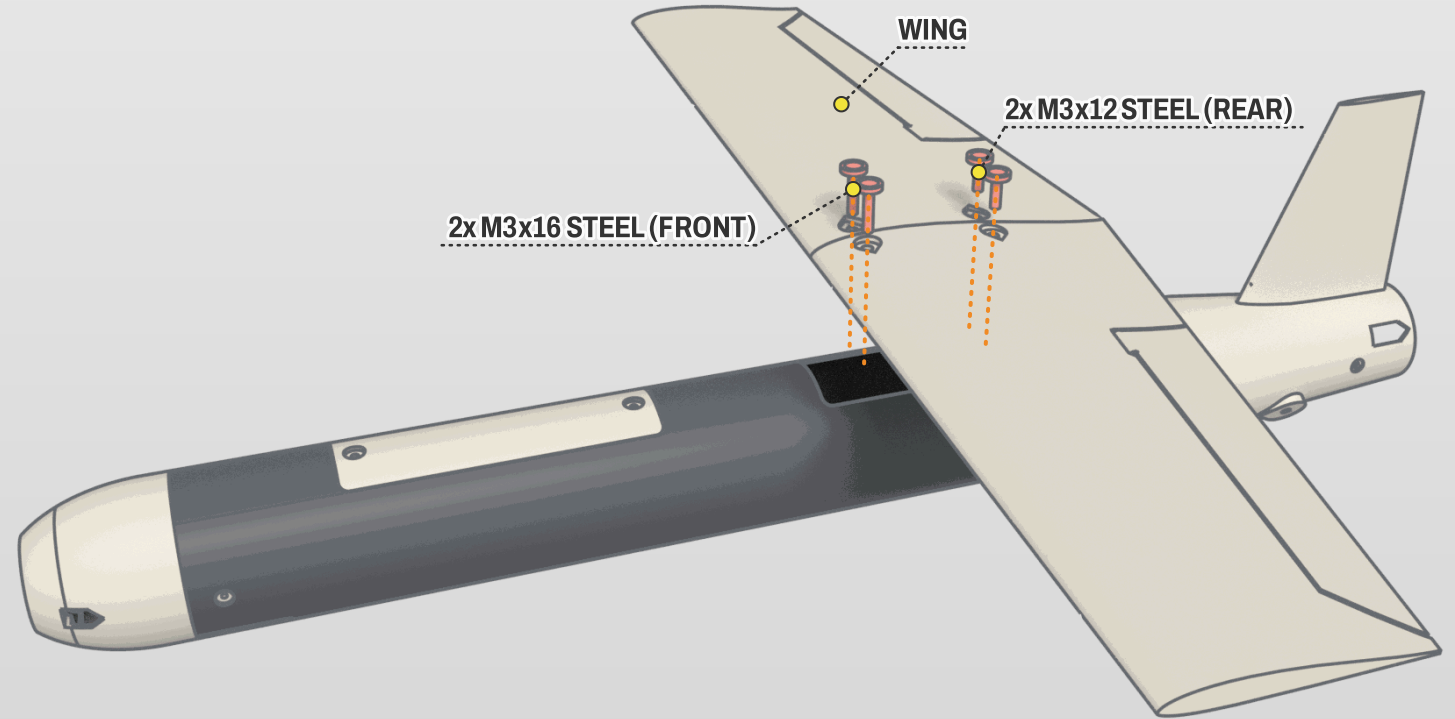
M2 × 5 SCREWS



FINAL ASSEMBLY



Glue the printed window filler (WF) under the wing root (epoxy), centred, with the wing on the tube so it lines up with the window. Push an M3 nut up into each of the bonded nut bars' pockets (two per bar). Set the wing into the tube's rectangular window (the filler plugs in) and bolt it with the steel countersunk screws in steel cup washers (flush in the recesses), two at each station, one in each wing half (each half is clamped on its own, off the glued seam): M3 × 16 at the front, M3 × 12 at the rear; they pull the nut bars up against the tube from inside. Snug them with a small driver, do not overtighten (the nut bars are printed).



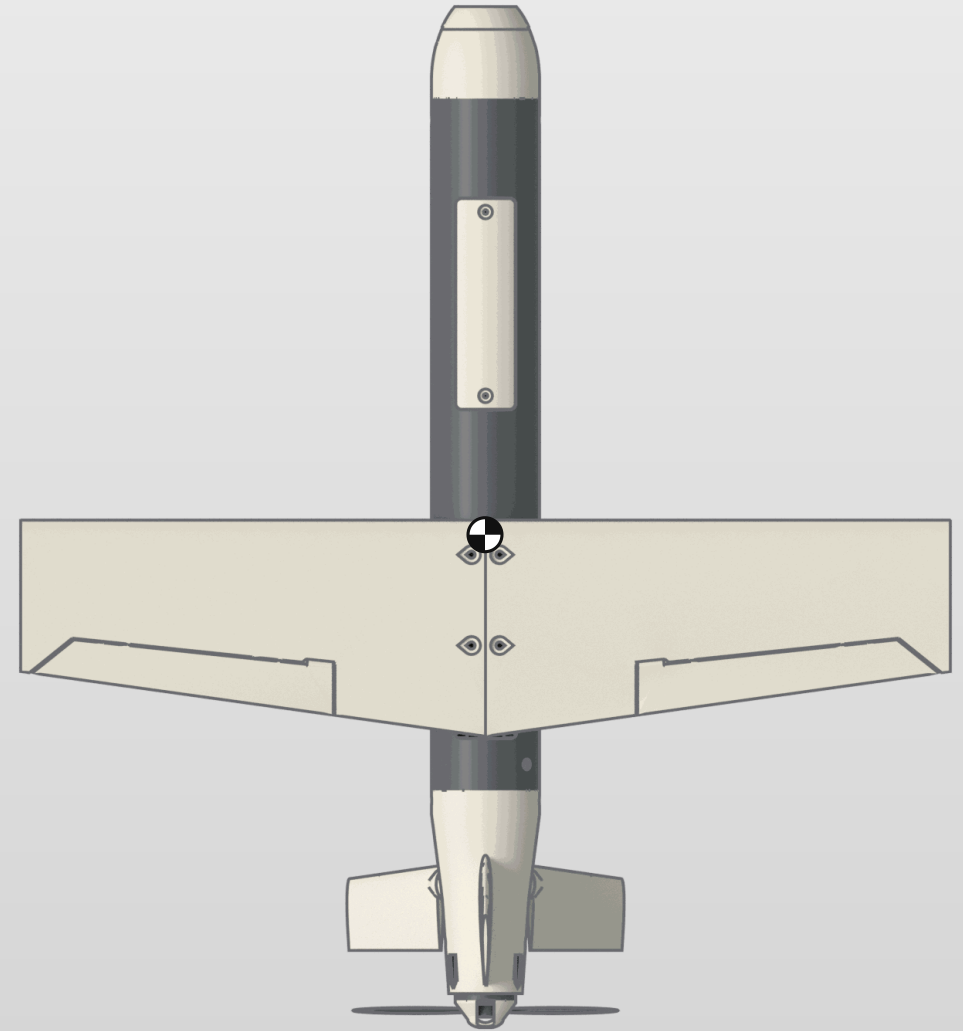
STEEL SCREWS

FINISHING BUILD



Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws $\pm 10^\circ$ on the tail panels, about $\pm 10^\circ$ (± 3.5 mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

Balance: CG 7 mm behind the wing leading edge at the root (mark it on the wing underside). Set the packs at the balance tick; for the first flight 2 mm further forward. If a heavy camera needs it, use the nose ballast cup.

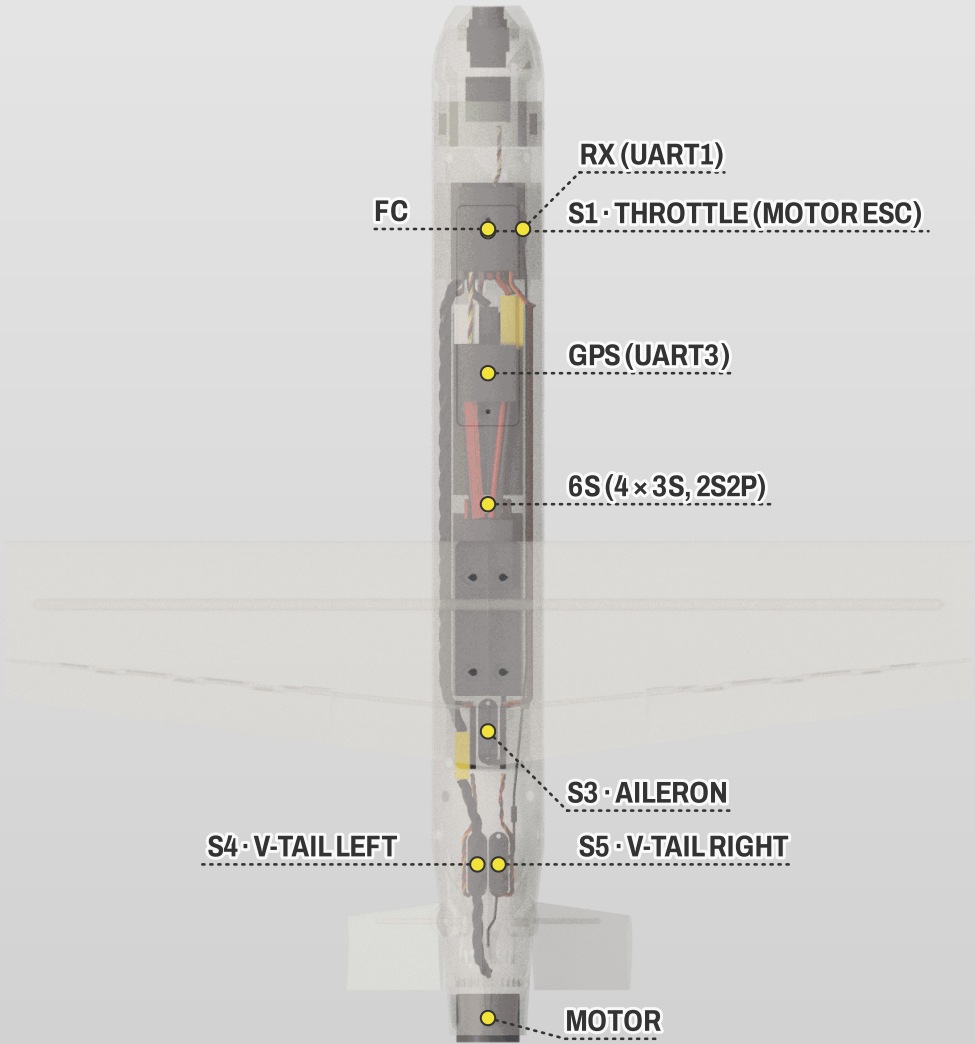


WIRING DIAGRAM



The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the package has this diagram as a separate large page.

PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	ESC (DShot300, bidirectional)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, tail module)
S5	V-tail right	Right tail-panel servo (ES9052MD, tail module)



ELECTRICAL INTEGRATION

Battery → FC power board (80 A current sensor) → ESC (motor on M1) → MR30 at the tail joint → motor. Main DC path 14 AWG (59 A on a pass, 65 A static), motor phases 16 AWG (48 A RMS). Cable lengths are measured on the 3D harness model; the cut list adds the termination allowances and slack. Full detail, pinouts, settings and tests: the **Electrical Integration manual** (PDF in the package and on the web page).

ID	CABLE	FROM	TO	AWG	CUT mm
W00	Pack leads and adapters	Each pack's own XT30 lead and 3S balance lead (front...	Adapter block on the rear pair: 2 x XT30 series adapters...	18	stock
W01a	Main battery lead	Parallel adapter output (XT30 male)	Charge / arming port: XT30 female, parked on the front...	14	110
W01b	6S balance lead	6S parallel balance board output	Charge port: 7-pin JST-XH, parked beside the XT30	26	110
W01	Battery lead (ESC side of the port)	XT30 male: plugs into the port to fly (the charger's...	FC power board BAT+ / BAT- pads	14	40
W02	ESC feed	FC power board ESC+ / ESC- pads	HDZero Halo 4in1 70A ESC AM32 BAT+ / BAT- pads (XT60...	14	35
W03	Motor phases, ESC side	HDZero Halo 4in1 70A ESC AM32 M1 pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the...	16	250
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat...	16	140
W05	ESC signal	HDZero Halo 4in1 70A ESC AM32 8-pin SH1.0 lead (only M1,...	FC S1 header (signal, GND); TLM to the R4 pad	28	45
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on edge on...	FC receiver header (5V, GND, RX1, TX1)	28	25
W07	ELRS antenna coax	RP1 V2 u.FL	Dipole feed at the centre of the fin dipole (z 39.5)	coax	385
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC GPS interface with compass (4V5, GND, RX3, TX3, SDA,...	28	65
W09	Video trunk (NOM nose)	Nose joint plug (camera on the nose; VTX on the nose or,...	FC pads: Vv, G, C1 (camera in), VTX (video out)	28	90
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	26	stock

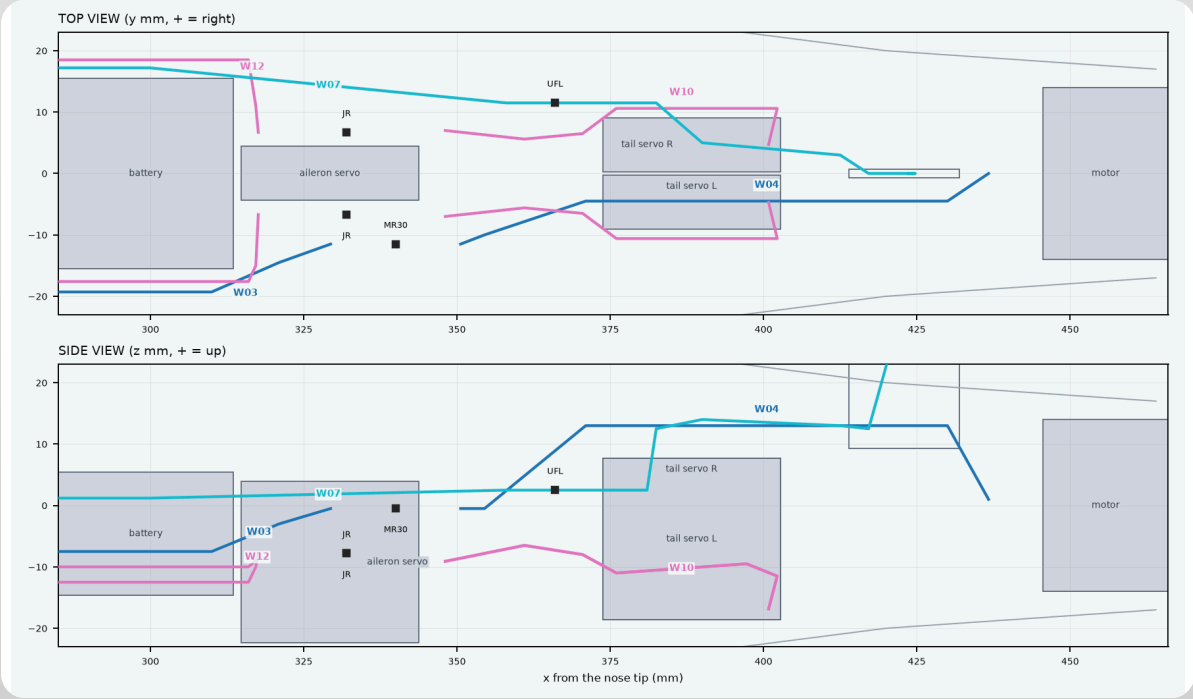
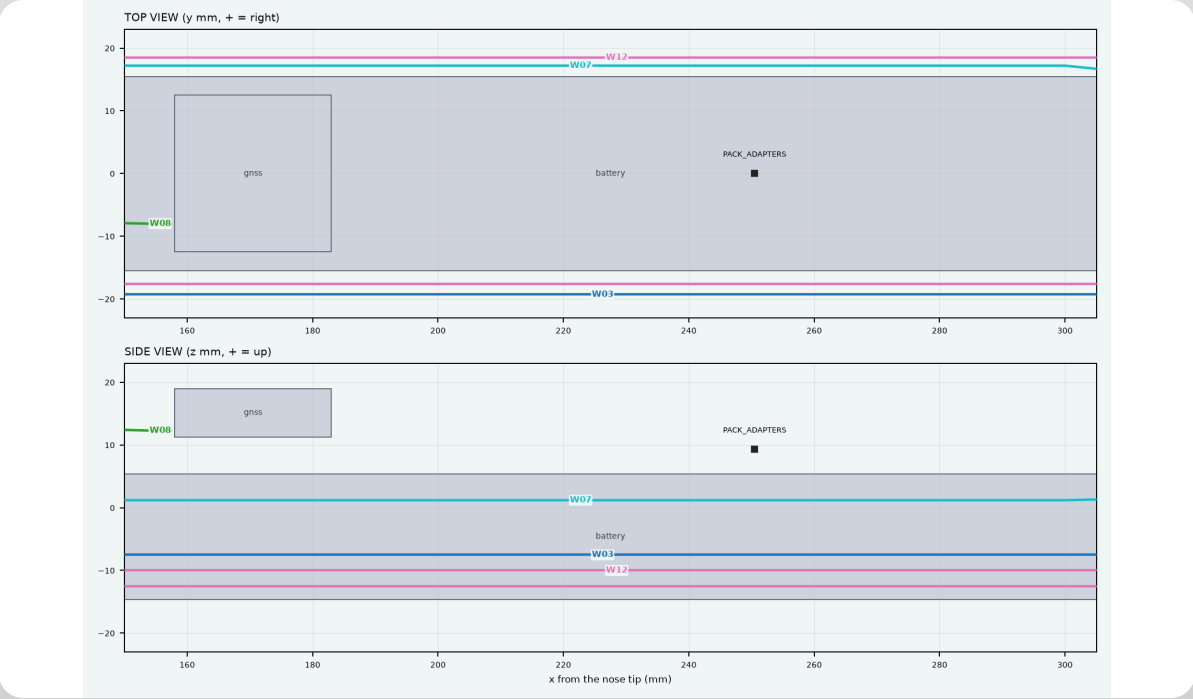
ELECTRICAL INTEGRATION (CONTINUED)

ID	CABLE	FROM	TO	AWG	CUT mm
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	26	stock
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	26	stock
W13	USB-C extension	FC USB-C port	USB-C female socket, parked between the charge plugs...	coax	stock

ELECTRICAL: POWER AND ROUTING

CABLE	CIRCUIT	CURRENT A (PASS / STATIC)	WIRE	DROP V	dT K / PASS	CONNECTOR RATING
W01+W02	Battery to ESC (DC)	59 / 65	14 AWG	0.10 (0.5 %)	33	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	48 / 53	16 AWG	0.39 (1.8 %)	55	MR30: 30 A cont / 40 A for 1 min per contact

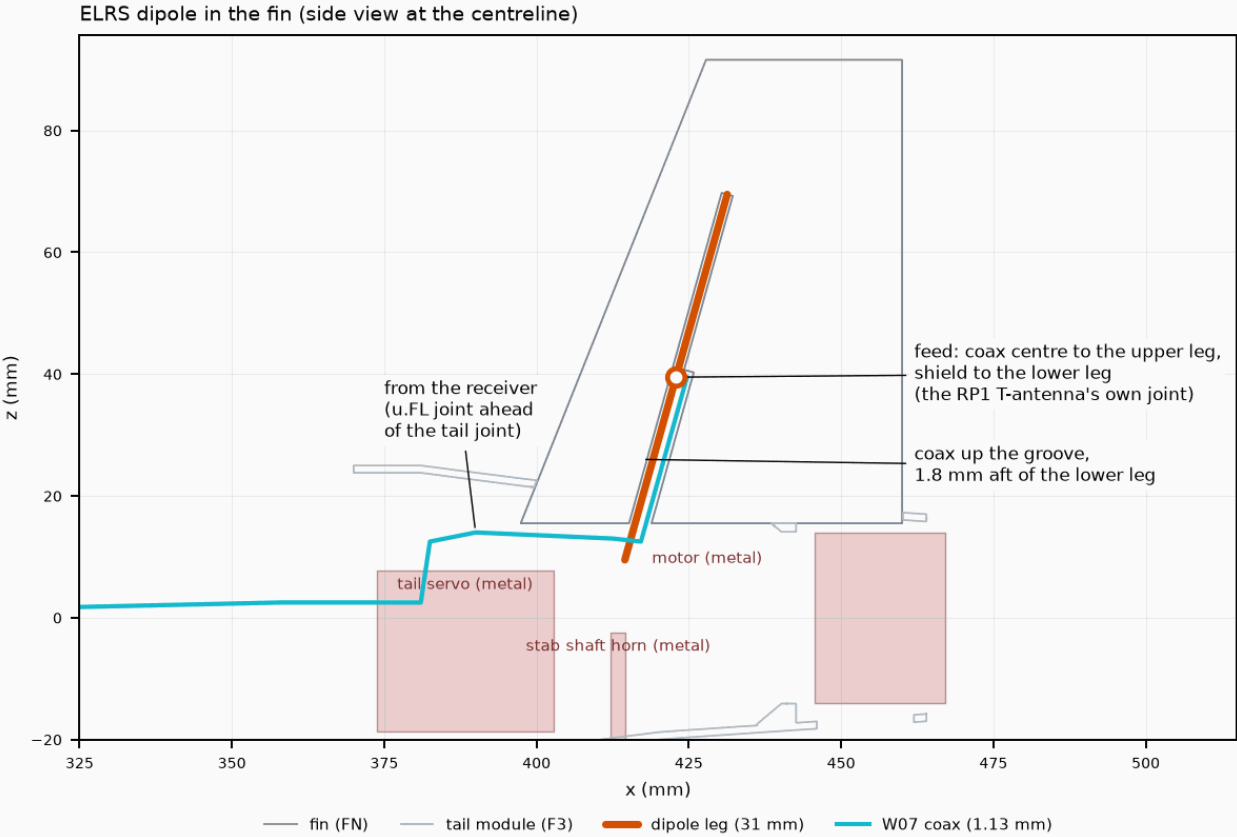
XT30: rated 15 A continuous / 30 A peak. Flown as the test base (70 % throttle cap, no governor) the pack draws about 22 A: fine. Only with the governor fitted (59–65 A) does it need an XT60 lead and a ready-made XT60 pack. Never connect the ESC lead's VBAT pin to the FC's servo header or a 5 V pad. Power-board jumpers: servo BEC 6 V; Vv battery voltage only if the camera and VTX accept 6S.



ELECTRICAL: RECEIVER ANTENNA



The receiver's antenna is a vertical dipole inside the fin, clear of all metal: the RP1 V2 T-antenna's two 31 mm legs lie in a groove on the fin's 30 % chord line with the T's joint (the feed) at the centre. The antenna's own coax runs down a parallel groove beside the lower leg into the tail module, to an in-line u.FL joint just ahead of the tail joint; a 1.13 mm u.FL extension continues to the receiver on the ESC. Unplug the u.FL joint with the MR30 and the servo plugs before pulling the tail module. Lay antenna and coax in the grooves before the fin is glued; three dots of epoxy, tape over the grooves.



ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 120B on the SpeedyBee F405 WING MINI (firmware target SpeedyBeeF405WING, ArduPlane 4.5 or later). Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), then work through the bench checks.

Flight controller: on the wing-bay floor, arrow forward; servo BEC jumper set to 6 V for the ES9052MD servos.

Firmware: ArduPlane 4.5 or later · **Board:** SpeedyBee F405 WING MINI (ArduPilot target: SpeedyBeeF405WING)

The setup document and the starting .param file are in the package (ArduPilot folder). Verify every value on the bench before flight.



FLIGHT TEST AND SPEED STAGES



The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay ≤ 0.2 mm (≤ 0.05 mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	23 %	23 m/s
35 m/s	26 %	28 m/s
40 m/s	29 %	32 m/s
45 m/s	33 %	37 m/s
50 m/s	36 %	41 m/s
55 m/s	40 %	45 m/s
60 m/s	44 %	50 m/s
65 m/s	47 %	53 m/s
70 m/s	51 %	58 m/s
75 m/s	55 %	62 m/s
80 m/s	59 %	66 m/s
85 m/s	63 %	70 m/s
90 m/s	67 %	74 m/s

SKYG

SKYG SPEED 120B

